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Review

Adoption and use of digital financial services: A meta analysis of barriers and facilitators

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ABSTRACT

The study of digital financial services is not recent; however, since COVID-19, new attention has been given to solutions that avoid face-to-face interactions. Accordingly, the number of studies on the topic of digital financial services has increased, covering new topics, such as digital wallets. Thus, this study presents a weight and meta-analysis that synthesises previous literature on digital financial services use. Using 121 articles, four analyses were performed for all financial solutions, but also per main service, such as digital banking, digital management and payment services, and digital wallets. Results show the relevance of technology adoption factors such as perceived usefulness, ease of use, but also security and trust dimensions for individuals to use these solutions. Additionally, results from the moderation analysis show that more money-oriented cultures tend to use these services more when perceiving them as easy to use. The findings of this study support practitioners and future research.

Introduction

Nowadays, the presence of mobile and/or digital solutions in individuals' lives is unquestionable (Baptista & Oliveira, 2016). Especially since the occurrence of the Covid-19 pandemic, people have given preference to digital solutions that require less human contact (Al-Qudah et al., 2022). One of the main affected areas was finance/payment services. These types of services are increasingly shifting to a digital environment, providing an easier way to manage payments, control finances, and proceed with payments, among other functionalities (Kaur et al., 2020). These services bring advantages not only for their users but also to banks, financial institutions, and almost any organisation that operates with digital payments (Chawla & Joshi, 2019). In this work, digital financial services are studied overall and per main area. In fact, these services might be divided into three main fields: digital banking, digital management and payment services, and digital wallets.

Digital banking is defined as a financial service made through mobile or digital means, allowing to perform transactions, trading, visualisation of history, and sometimes permitting advisory services and cross-selling products (Baptista & Oliveira, 2015). On the other hand, digital management and payment services are not implicitly connected to a specific bank since FinTechs usually offer them. These solutions allow users to invest their money, provide investment proposals, define financial goals, perform payments, and present visualisations of history and performance (Gerlach & Lutz, 2021). Finally, digital wallets are defined as mobile applications that work as a wallet in a way that allows performing payments and storing information such as credit cards, passport details, several payment accounts, passwords, etc. (George & Sunny, 2022).

The number of studies on these services has increased greatly in the last few years, mainly covering the three areas mentioned above. These results reflect the maturity of digital financial services, especially digital banking. Given that the literature on the topic is somewhat scattered and the research process has become complex, there is a need to provide a comprehensive and synthesised picture of the progress in the field of digital financial services by evaluating and integrating the findings described in the literature, overall and per service, especially due to the emergence of more recent digital financial services, like digital wallets. Therefore, it is now relevant to understand what literature has investigated on antecedents of use, over the years, and understand how these factors might change across services. Hence, this study intends to answer the following research questions: (1) What are the main motivators and

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barriers influencing individuals' intention to use digital financial services?; (2) How do these factors vary across different digital financial services?; (3) How do these factors vary across different socio-demographic contexts?

To answer the above questions (1) and (2), a weight and meta-analysis will be performed for all digital financial services and an analysis per main service: digital banking, digital management and payment services, and digital wallets. Moreover, a moderation analysis will be developed in order to compare results across different socio-demographic contexts, providing an answer to the research question (3). A meta and weight analysis is considered a stronger method to integrate quantitative results than the usual literature reviews (Schmidt & Hunter, 2016), being a robust method that provides a concise understanding of a topic, uncovering relationships, disparities in results, and gaps (Urbach et al., 2009). Thus, one of the main reasons to use meta-analysis is its capacity to compare effect sizes across different effects. Besides the meta-analysis, a weight analysis is also performed, allowing the identification of the most frequently used constructs (Jeyaraj et al., 2006). The combination of a meta and weight analysis provides one of the most comprehensive analysis available (Baptista & Oliveira, 2016). Overall, these techniques together allow us to summarise the different studies, identify the most important variables, and quantify the moderating effects. Therefore, we believe these are strong methods that allow to answer the proposed research questions.

The article makes the following contributions. First, to the best of our knowledge, this is the first time that a weight and meta-analysis has been performed for this whole set of digital financial services, especially regarding digital management and payment services and digital wallets, providing an overall perspective of the status of digital financial services, but also a discriminated view of each major service. Secondly, it will contribute to a better understanding of the most-used variables and theories, facilitating theory development in the fields of information systems and financial research. Thus, researchers can make use of these findings to help in the selection of factors to analyse the use of digital financial services. Finally, creating a meta and weight analysis per service will allow an understanding of how digital financial services may differ between them in terms of main motivators and barriers.

The paper is structured as follows. The next section presents the research methodology with a description of the selected studies and the merging process. Section 3 presents the results, including some descriptive statistics, the meta and weight analysis results, and the moderator's analysis. Section 4 introduces the discussion of results and their implications. Section 5 presents the limitations and future research directions, followed by the conclusion in Section 6.

Literature review

There is a broad set of digital financial services, according to what they allow to perform, which organisations are behind the service, and which functionalities are available, among others. Three primary services can be identified: digital banking, digital management and payment services, and digital wallets. Digital banking is the most common digital financial service. It is usually associated with a banking organisation and allows banking activities to be performed either electronically or online (Al-Dmour et al., 2019). These banking services are available at any time, allowing tasks to be executed anytime and anywhere, avoiding queues, and reducing operating costs (Inder et al., 2022). Among the functions available, digital banking enables the performance of various transactions, trading, and visualisation of history and sometimes permits access to advisory services and cross-selling products (Baptista & Oliveira, 2015). Overall, these services can be boosted with the implementation of emerging technologies, such as adaptative business intelligence (Arjun et al., 2021). However, although this might present benefits, the technology acceptance depends on a variety of factors, especially security and convenience ones. On the other hand, although digital management and payment services perform many similar activities to those of digital banking, they are usually offered by FinTechs (Mainardes, Costa, & Nossa, 2023) or e-commerce organisations like Amazon (Jena, 2022). These services allow users to invest money, provide investment proposals, define financial goals, make payments, and present visualisations of history and performance (Gerlach & Lutz, 2021). In fact, due to the high amounts of data collected and available, several advanced models of deep learning or reinforcement learning are being developed to support decision making (Singh et al., 2022). Finally, digital wallets are the newest form of digital financial service. These are defined as mobile applications that substitute real wallets, allowing users to make payments without the participation of financial intermediaries (Tran Le Na & Hien, 2021) and store information, such as credit cards, passport details, several payment accounts, passwords, etc. (George & Sunny, 2022). In fact, digital wallets are increasingly being investigated, and not only for financial purposes. For example, blockchain based wallets for health information are also being studied (Maher et al., 2023; Mittal, Gupta, Chaturvedi, Chansarkar, & Gupta, 2021). Overall, the maturity of the digital financial services is clear, however, there are clear opportunities due to the digital advancements, like advanced methods and blockchain. Given this, it is important to understand what literature has investigated on this topic.

Research methodology

Methods

A meta-analysis allows one to summarise the quantitative findings from prior research on similar topics. Thus, one of the main reasons to use the meta-analysis is its capacity to compare effect sizes across different effects (Schmidt & Hunter, 2016). Although having the limitation of only evaluating quantitative studies, the number of quantitative articles on the topic of digital finance services is strongly increasing, being therefore especially relevant nowadays to provide this analysis. Thus, despite some variable criticism (Borenstein et al., 2009), some researchers consider that a meta-analysis is more robust when compared to descriptive literature reviews, creating a consolidated view of the research on a topic (Schmidt & Hunter, 2016). Given this, we believe that meta-analysis is appropriate for our study. Meta-analyses allow researchers to compute a pooled estimate of the explanatory variables used to explain the target ones, combining different effect sizes from various studies. This meta-analysis followed the standard meta-analytical calculation procedures (Sampaio et al., 2017). This method requires the selection of quantitative studies that present the following metrics: sample size, and correlation coefficients. These measures are needed to compute the pooled estimate of the relationships studied. After the collection of studies, it is necessary to merge all constructs whose definition is the same but might have different names. This process will also allow us to understand the most used dependent variable, as well as the most tested relationships. Having this, it will then be possible to compute the meta-analysis.

Additionally, a weight analysis was conducted, as it enables estimating an independent variable's importance, indicating its predictive power to the target variable (Jeyaraj et al., 2006). The combination of a meta and weight analysis provides one of the most comprehensive analysis available (Baptista & Oliveira, 2016). After data collection, the weight is calculated by dividing the number of times the independent variable is significant by the total number of relationships between the independent variable and a dependent one. Each dependent variable will therefore be classified according to its weight. The next sub-section presents the steps described above to conduct the meta and weight analysis, following the structure of previous meta-analysis investigations (Naranjo Zolotov et al., 2018).

Selection of studies

This study followed a meta-analytical study approach (Schmidt &

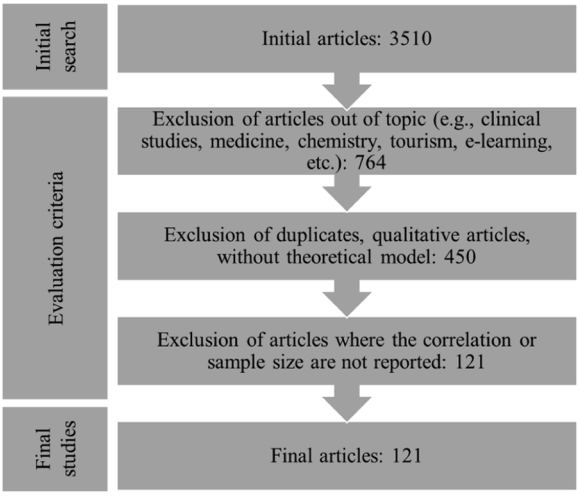


Fig. 1. Studies selection process.

Table 1
Moderators description.

Variable	Description	Source
Human development index (HDI)	Index that comprises three indicators of a country: standard of living, life expectancy, and years of schooling	(Nations, 2023)
Global innovation index (GII)	Is composed of 81 indicators measuring the political environment, education, knowledge creation and innovation level of each country	(Organization, 2022)
Uncertainty avoidance	Refers to the degree to which people feel intimidated by uncertain situations	(Hofstede & Minkov, 2010)
Masculinity	Refers to the degree to which dominant responsibilities/power is for masculine/feminine individuals	
Individualism	Represents the extent to which people value social or personal needs	
Long-term orientation	Represents the degree to which people value short term to long term fulfilment, it also refers to the type of values people are oriented by in terms of traditional values vs innovative/ recent ones;	
Power distance	Represents the extent to which people feel that power is on institutions without consultation of lower-level entities	
Indulgence	Measures the extent to which an individual feels to have and allow themselves to have an enjoyable and desired life	

Hunter, 2016). In particular, we applied a bivariate meta-analysis to test direct effects and meta-regressions to analyse the possible moderators. We detail each step of our meta-analysis below. Studies were collected during the month of November 2022 from the most well-known databases, such as Scopus, Science Direct and Web of Science (Webster & Watson, 2002). Queries were built on those databases using the fields available, the logical operators (AND/OR) and the most relevant keywords. First, to guarantee that quantitative articles were found, keywords like questionnaire, survey, regression, or structural equation modelling were used. Then, to define the context of the studies, keywords like digital banking, digital payment, digital wallet, and digital assistant were used. Other variations of the keywords were also searched, such as mobile/online banking, electronic wallet, and e-wallet. Finally, to capture the specific adoption and use behaviour, keywords like intention to adopt, intention to use, adoption, and use were also added. From the queries, 3510 initial articles were found. A rigorous analysis was made on these, based on a set of criteria: (1) studies published under a peer-review process; (2) studies on the scope of the analysis (excluding specific contexts, such as clinical studies, medicine, tourism, e-learning, etc.); (3) quantitative studies that show correlation values and sample sizes; (4) independent datasets; (5) studies using datasets already included were removed to avoid study bias by using the same sample (Wood, 2008). These criteria was established to make sure only quantitative peer-reviewed studies on the topic were selected, following the recommendations of previous recent meta-analysis (Franque et al., 2021; Neves et al., 2022). After this screening process, 121 articles met all criteria, being able to be used in the meta and weight analysis. Several pieces of information were recorded from each article, namely the publication year, source, independent variable, dependent variable, the correlation coefficient of each relationship, method, type of technology, sample size, country, and journal. Fig. 1 describes the selection process.

Variables merger and relationships

At the onset of the analyses, the main information to be extracted from the articles is the relationships between dependent and independent variables. Nevertheless, the occurrence of variables with different names but the same meaning is recurrent. For example, performance expectancy and perceived usefulness have been identified as holding the same meaning. Therefore, before starting the analysis, it is crucial to merge variables into one single name when representing the same concept. This process was conducted for both dependent and independent variables. The coding scheme and data extraction procedures followed previous research in meta-analyses (Rosario et al., 2016). After deciding the coding classification criteria, two research assistants coded the effect sizes. A third judge resolved the disagreements. After this process, 384 relationships were identified; however, only 30 could be used, given that only relationships that occurred three or more times can be used to perform the meta-analysis (Baptista & Oliveira, 2016). From the selected relationships, the main dependent variables identified were intention to use, use behaviour, intention to continue using, attitude and satisfaction. The variables codification is presented in Appendix A.

Moderation analysis

When testing for the significance of independent variables, it is also possible to evaluate moderation effects. This is especially relevant to understand how the factors that influence individuals intention to use digital financial services vary across different socio-demographic contexts. Therefore, several variables were collected as possible moderators of relationships. Table 1 presents the collected variables to be tested as moderators, as well as a short description. Some socio-economic variables were selected, such as the human development index (HDI) and the global innovation index (GII). HDI is a measure that summarises the standard of living, life expectancy and education and measures the overall development of a country. GII comprises 81 indicators and measures the economic innovation level of a country. Overall, it is expected that countries with higher levels of HDI and GII will present higher levels of use of digital financial solutions, given the solid digital component of these solutions. Also, more developed economies tend to invest more in the development of these solutions in order to facilitate payments and money investments (Hassan & Wood, 2020). Moreover, Hofstede's cultural indicators were collected (Hofstede & Minkov, 2010), given the substantial impact cultures might have on technology adoption and use (Tam & Oliveira, 2017). Hence, uncertainty avoidance, masculinity, individualism, long-term orientation, power distance and indulgence were collected.

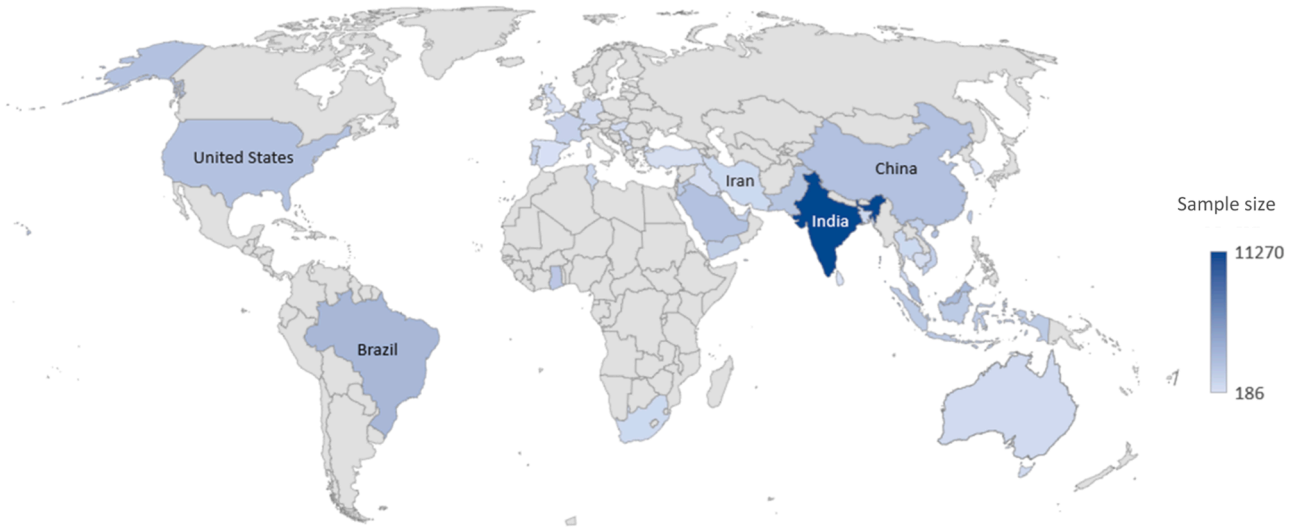


Fig. 2. Sample distribution by country.

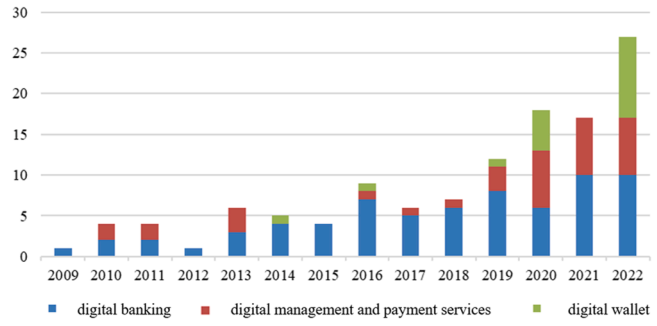


Fig. 3. Selected studies by year.

Results

This section is composed of a descriptive analysis of the articles found and analysed. This includes ana analysis of the countries where studies have been conducted, the type of digital service, and the journals in which the topic of digital financial services have been published. Then, the results of the meta and weight analysis are presented, respectively. On this sub-section, four meta and weight analysis results are presented on overall digital financial services, and then per service, respectively – digital banking, digital payment assistants, and digital wallets. Finally, the moderation analysis results are described. This last analysis was only conducted for all digital financial services, and not individually per service, given that moderation analysis can only be significant when working with more than twenty relationships (Geyskens et al., 2009).

Descriptive analytics

From the 121 datasets, samples covered 37 countries. Fig. 2 presents the cumulative sample size per country, showing the top three countries as India (22% of the respondents), Brazil (6% of the respondents) and China (5% of the respondents). Regarding the research years, Fig. 3 summarises the distribution of studies over the years, categorised by type of digital financial solution. The selected studies range from 2009 to 2022, presenting an increased tendency over the years since the last three years represent 51% of the studies. Moreover, the increasing interest in digital wallet research is noticeable, given its strong presence in recent years.

Regarding the source of the articles, Table 2 summarises the number of articles per journal. As described, articles mainly belong to the finance, management, and information systems fields, especially to the following journals: international journal of bank marketing (25%), journal of retailing and consumer services (6%), and international journal of information management (5%). Finally, in terms of the main areas of digital financial services, the majority of articles are about digital banking (69 studies), followed by digital management and payment services (34 studies), and lastly, digital wallets (18 studies). In fact, digital wallets are being increasingly studied, showing great prominence in recent studies in the digital financial arena.

Weight and meta-analysis results

Following the recommendations from recent meta-analyses, we only included studies that reported the correlation effects in our analysis, reducing potential conversion biases (Roth et al., 2018). Collecting a set of quantitative measures is required to calculate this, namely the statistically significant and non-significant relationships, the effect size, and the sample size (Cook, 1991). The meta-analytical effect sizes were corrected by sample size (Hunter & Schmidt, 2004). The random-effects model was used instead of the fixed-effects one, given the high variance in terms of effect sizes and the heterogeneity between studies due to different samples (Borenstein et al., 2010). This method considers within and between variance between studies, while the fixed effect model only considers the variation within studies. Given this, the random effect is a much more realistic method and widely used on meta-analysis (Franque et al., 2021; Neves et al., 2022). We also evaluated the publication bias, which refers to the possible tendency for studies with significant and positive results to get published instead of the studies with no statistically significant or negative results. This bias can influence the meta-analysis results. Therefore, Cochran's Q, I^2 heterogeneity measure, failsafe number and Egger's test were evaluated. In Cochran's Q, the heterogeneity is proven by the significance level. The I^2 presents heterogeneity in a range from 0% to 100%. We also assessed the failsafe number (FNS) (Rosenthal & Rubin, 1991). The failsafe number comprises the number of non-significant or unpublished studies that would be necessary to refute the findings of each relationship tested. Finally, the effect size was also checked for evidence of publication bias using Egger's test (Egger et al., 1997), where we evaluated whether the data distribution was a representative sample rather than asymmetric. Additionally, we performed meta-regression in order to analyse possible moderators in the main relationships tested. The analyses were conducted using packages of R (version 4.0.2).

In order to provide a comprehensive view of digital financial

Table 2
Selected studies per journal.

Journal	N° articles	Source
Asia pacific journal of marketing and logistics	1	(Lisana & Handarkho, 2022)
Banks and Bank Systems	1	(Akhter et al., 2022)
Cogent Business & Management	4	(Aji et al., 2020; Ly et al., 2022; Nguyen-Viet & Ngoc Huynh, 2021; Ojo, Fawehinmi, Ojo, Arasanmi, & Tan, 2022)
Computers and industrial engineering	1	(Kapoor et al., 2022)
Computers in Human Behavior	5	(Baptista & Oliveira, 2015; Kim et al., 2010; Malaquias & Hwang, 2016; Tam & Oliveira, 2016; Tao Zhou et al., 2010)
Computers in Human Behavior Reports	1	(Ly & Ly, 2022)
Decision Support Systems	1	(Zhou, 2013)
Digital Business	1	(Alnemer, 2022)
Digital policy, regulation and governance	1	(Adjei et al., 2020)
Electronic Commerce Research	1	(Agárdi & Alt, 2022; Nathan et al., 2022)
Electronic Commerce Research and Applications	1	(Verkijika, 2020)
Frontiers in psychology	1	(Almaiah et al., 2022)
Global business review	1	(Khan et al., 2021)
Information & Management	1	(Tam & Oliveira, 2019)
Information development	4	(Ahmad et al., 2020; Rahman et al., 2022; Yuan et al., 2016; Zhou, 2011b)
Information Systems and e-Business Management	1	(Liu et al., 2019)
Information technology for development	1	(Pal et al., 2020)
Innovation and Management Review	1	(Abegao Neto & Figueiredo, 2022)
Innovative marketing	1	(Al-Hattami et al., 2021)
International Journal of Bank Marketing	30	(Al-Ajam & Md Nor, 2015; Anouze & Alamro, 2020; Baabdullah et al., 2019; Banerjee & Sreejesh, 2022; Bankuoru Egala et al., 2021; Bitrián et al., 2021; Chaouali et al., 2019; Chaouali & El Hedhli, 2019; Chauhan et al., 2019; Chawla & Joshi, 2019; Chemingui & Lallouna, 2013; Chong et al., 2010; Elhajjar & Ouaida, 2020; Farah et al., 2018; Garzaro et al., 2021; Glavee-Geo et al., 2017; Gupta & Arora, 2017; Humbani & Wiese, 2019; Idoga et al., 2018; Kamboj et al., 2022; Kesharwani & Bisht, 2012; Lifen Zhao et al., 2010; Mortimer et al., 2015; Mzoughi & M'Sallem, 2013; Namahoot & Laohavichien, 2018; Poromatikul et al., 2020; Rafdinal & Senalasari, 2021; Rejman Petrović et al., 2022; Riquelme & Rios, 2010; Shankar & Jebarajakirthy, 2019)
International Journal of Data and Network Science	3	(Al-Zu'bi & Al-Gasawneh, 2022; Astari, Yasa, Sukaatmadja, & Giantari, 2022; Rahadi et al., 2022)
International Journal of E-Services and Mobile Applications	1	(Ibrahim et al., 2022)
International journal of finance and economics	1	(Mohd Thas Thaker et al., 2022)
International journal of human-computer studies	1	(Gupta et al., 2022)
International Journal of Information Management	6	(Alalwan et al., 2017; Baabdullah et al., 2019; Leong et al., 2020; Lin, 2011; Oliveira et al., 2014; Nidhi Singh, Sinha, & Liébana-Cabanillas, 2020)
Internet research	3	(Baptista & Oliveira, 2017; Teoh et al., 2013; Zhou, 2011a)
Journal of African business	1	(Agyei et al., 2022)
Journal of Asia business studies	1	(Kaur & Arora, 2021)
Journal of business research	1	(Ciunova-Shuleska et al., 2022)
Journal of computer information systems	1	(Chen & Li, 2017)
Journal of Economics and Business	1	(Gerlach & Lutz, 2021)
Journal of Financial Services Marketing	2	(Abikari et al., 2022; George & Sunny, 2022)
Journal of High Technology Management Research	1	(Aboobucker & Bao, 2018)
Journal of Indian business research	1	(Madan & Yadav, 2016)
Journal of Internet Commerce	1	(Al Khasawneh, 2015)
Journal of Marketing Analytics	1	(Martinez & McAndrews, 2022)
Journal of open innovation	1	(Kumar et al., 2020)
Journal of Organizational Computing and Electronic Commerce	1	(Changchit et al., 2017)
Journal of promotion management	1	(Khoa, 2020)
Journal of retailing and consumer services	7	(Al-Sabaawi et al., 2021; Jebarajakirthy & Shankar, 2021; Kaur et al., 2020; Park et al., 2019; Shaw, 2014; Singh & Sinha, 2020; Zhou et al., 2021)
Journal of science and technology policy management	4	(Inan et al., 2021; Rahi & Abd. Ghani, 2019; Sharma & Khurana, 2022)
Journal of services marketing	1	(Bailey et al., 2022)
journal of strategic marketing	1	(Roy et al., 2017)
Managing service quality	1	(Chen, 2013)
Marketing intelligence and planning	4	(Akturan & Tezcan, 2012; Deb & Lomo-David, 2014; Jaiswal et al., 2022; Kaur et al., 2020; Ofori et al., 2017)
Mobile information systems	1	(Hidayat-Ur-Rehman et al., 2021)
Research policy	1	(Pal et al., 2021)
Review of international business and strategy	1	(Alshurideh et al., 2021)
South Asian journal of business studies	1	(Gupta & Arora, 2020)
Technology in society	6	(Abbasi et al., 2022; Al-Okaily et al., 2020, 2022; Ho, Wu, Lee, & Pham, 2020; Khan, 2022; Lian & Li, 2021; Rahi et al., 2019)
Telematics and Informatics	3	(Afshan & Sharif, 2016; Hanafizadeh et al., 2014; Mehrad & Mohammadi, 2017)
The Service Industries Journal	1	(Koenig-Lewis et al., 2015)

services, an overall model was estimated, including all services, as well as three individual ones for each solution: digital banking, digital management and payment services, and digital wallets. Starting with the main model, Table 3 presents the meta-analysis results. For the main model, four target variables were identified: intention to use, intention to continue to use, attitude, and perceived usefulness. The first columns present the number of times the relationship was examined (O), the cumulative sample size (N), the correlation found in the studies corrected by sample size (r), and the confidence interval (lower and upper bounds) of 95% confidence level. Regarding this last aspect, one can consider the relationship as not statistically significant only when the interval includes the value zero. Otherwise, they are considered statistically significant. Finally, regarding heterogeneity, the Q Cochran measure and the scale-free index of heterogeneity (I2) are presented, indicating whether the data refute the homogeneity hypotheses and the percentage of variance in the dataset resulting from heterogeneity (Higgins & Thompson, 2002), respectively. This structure is present in

Table 3
Meta and weight analysis results for all digital financial services.

Dependent variable	Independent variable	*O*	*N*	*r*	CI (95%)		Q	I^2	*FNS*	*Egger's Intercep (sig)*	*Weight*	*Type*
Intention to use	Perceived usefulness	50	20,362	0.37	0.32	0.31	808.36*	0.94	79,140	0.36	0.98	BP
	Perceived ease of use	33	13,047	0.19	0.11	0.28	797.15*	0.96	9334	0.16	0.91	BP
	Subjective norms	31	13,964	0.17	0.10	0.24	526.32	0.94	4930	0.90	0.84	BP
	Trust	26	10,046	0.29	0.21	0.36	453.08*	0.95	11,200	0.09	0.96	BP
	Attitude	22	10,364	0.55	0.42	0.65	1562.45*	0.99	10,069	0.63	1.00	BP
	Facilitating conditions	21	8123	0.16	0.09	0.24	257.52*	0.92	1698	0.06	0.76	
	Perceived risk	19	9799	-0.08	-0.21	0.06	792.63*	0.98	1219	0.25	0.26	
	Hedonic motivations	14	5091	0.22	0.07	0.36	401.40*	0.97	2183	0.46	0.86	BP
	Price value	11	3773	0.38	0.07	0.62	992.10*	0.99	27,120	0.90	0.91	BP
	Habit	10	4307	0.36	0.23	0.47	176.65*	0.95	2374	0.34	0.90	BP
	Personal innovativeness	8	3005	0.22	0.14	0.29	30.27*	0.77	477	0.19	0.88	BP
	Perceived security	7	2439	0.21	0.11	0.31	39,76*	0.85	331	0.40	0.86	BP
	Convenience	7	3098	0.36	0.10	0.58	381.79*	0.98	3042	0.01	1.00	BP
	Compatibility	5	3023	0.57	0.14	0.82	697.46*	0.99	11,680	0.77	1.00	BP
	Awareness	3	1208	0.19	0.01	0.35	18.30*	0.89	57	0.35	1.00	PP
	Cost	3	837	-0.12	-0.35	0.12	25.56*	0.92	11	0.37	0.33	
	Privacy concern	3	1242	0.01	-0.22	0.23	31.44*	0.94	0	0.30	0.33	
Intention to continue to use	Satisfaction	12	4560	0.48	0.32	0.61	451.20*	0.98	12,810	0.71	1.00	BP
	Perceived usefulness	9	3846	0.25	0.16	0.34	68.25*	0.88	923	0.13	1.00	BP
	Trust	8	3240	0.26	0.17	0.34	44.72*	0.84	722	0.95	1.00	BP
	Perceived ease of use	6	2839	0.14	0.05	0.23	29.30*	0.83	121	0.71	0.83	BP
	Perceived risk	5	1986	-0.18	-0.25	-0.10	11.96*	0.67	120	0.09	0.00	
Attitude	Perceived usefulness	16	6676	0.39	0.34	0.44	93.00*	0.84	8532	0.19	1.00	BP
	Perceived ease of use	15	4749	0.28	0.17	0.38	223.74*	0.93	2732	0.56	0.93	BP
	Subjective norms	5	2107	0.35	0.12	0.54	120.63*	0.97	727	0.23	1.00	BP
	Perceived risk	4	2463	-0.39	-0.73	0.11	386.11*	0.99	1296	0.75	0.00	
	Compatibility	4	1510	0.37	0.28	0.44	9.39*	0.68	386	0.58	1.00	PP
	Benefits	4	1340	0.36	0.09	0.58	84.64*	0.97	373	0.61	0.75	
	Privacy concern	3	1344	0.10	-0.05	0.25	15.11*	0.87	15	0.82	0.67	
Perceived usefulness	Perceived ease of use	9	3907	0.55	0.46	0.62	112.02*	0.93	8635	0.74	1.00	BP

Note: (O) number of observations taken from the analysis of the studies; (N) number of accumulated samples of the assessed studies; r =correlation found in the studies correct by sample size; CI (95%)=confidence interval; Q= test of heterogeneity at the individual; I^2=scale-free index of heterogeneity; (*) = $p < 0.01$; ns= not significant; NC = not calculated; Egger's intercept = Asymmetry test.

Table 4
Meta and weight analysis results for digital banking services.

Dependent variable	Independent variable	*O*	*N*	*r*	CI (95%)		Q	I^2	*FNS*	*Egger's Intercep (sig)*	*Weight*	*Type*
Intention to use	Perceived usefulness	24	8087	0.41	0.33	0.49	416.93*	0.95	21,858	0.55	1.00	BP
	Subjective norms	19	7649	0.12	0.06	0.18	124.00*	0.86	792	0.93	0.79	
	Perceived ease of use	17	5512	0.18	0.11	0.25	98.79*	0.84	1215	0.44	0.88	BP
	Attitude	16	8148	0.58	0.42	0.70	1432.50*	0.99	75,378	0.74	1.00	BP
	Trust	10	4364	0.32	0.16	0.47	277.40*	0.97	2683	0.25	1.00	BP
	Perceived risk	9	3771	-0.04	-0.34	0.22	550.93*	0.99	216	0.06	0.20	
	Hedonic motivations	8	3254	0.28	0.07	0.47	281.15*	0.98	1614	0.72	0.78	
	Price value	7	2648	0.39	-0.06	0.71	917.22*	0.99	21,112	0.72	0.88	BP
	Convenience	6	2676	0.24	0.14	0.33	37.53*	0.87	390	0.83	0.86	BP
	Personal innovativeness	6	2268	0.23	0.15	0.30	18.38*	0.73	289	0.25	0.83	BP
Attitude	Perceived usefulness	12	5499	0.39	0.33	0.45	63.78*	0.83	5156	0.12	1.00	BP
	Perceived ease of use	10	3389	0.26	0.10	0.40	200.77*	0.96	1168	0.58	0.90	BP
	Perceived risk	4	2463	-0.39	-0.73	0.11	386.11*	0.99	1296	0.75	0.00	
	Compatibility	4	1510	0.37	0.28	0.44	9.39*	0.68	386	0.58	1.00	PP
	Subjective norms	4	1746	0.39	0.12	0.60	108.59*	0.97	630	0.13	1.00	PP
Satisfaction	Information quality	3	1282	0.21	0.13	0.29	4.36	0.54	66	0.05	1.00	PP
	System quality	3	1282	0.22	-0.06	0.47	53.08*	0.96	75	0.39	0.67	
	Perceived usefulness	3	1051	0.37	0.24	0.49	11.21*	0.82	212	0.19	1.00	PP
Intention to continue to use	Satisfaction	7	2618	0.40	0.24	0.53	124.75*	0.95	1732	0.64	1.00	BP
	Perceived usefulness	4	1574	0.29	0.11	0.45	43.05*	0.93	253	0.31	1.00	PP
	Trust	4	1476	0.29	0.12	0.44	35.72*	0.92	242	0.56	1.00	PP
Adoption	Perceived usefulness	5	2359	0.41	0.21	0.58	107.49*	0.96	976	0.41	1.00	BP
	Trust	4	1759	0.38	-0.04	0.68	208.59*	0.99	424	0.25	0.75	
	Perceived risk	3	1147	-0.20	-0.40	0.02	26.25*	0.92	71	0.33	0.00	
	Perceived ease of use	3	1573	0.36	0.19	0.51	21.99*	0.91	364	0.09	1.00	PP
	Intention to use	3	899	0.56	0.44	0.66	11.71*	0.83	657	0.06	1.00	PP
Use behaviour	Intention to use	5	1985	0.33	0.20	0.44	36.59*	0.89	535	0.73	1.00	BP
Perceived usefulness	Perceived ease of use	4	1650	0.58	0.39	0.72	77.61*	0.96	2195	0.83	1.00	PP

all the meta-analysis results. It can be observed that only one relationship did not reject the homogeneity hypothesis ($p<0.01$) (subjective norms to intention to use). Moreover, most relationships present a high heterogeneity level, above 90%, meaning that the variability among effect sizes is not caused by sampling error, but true heterogeneity between studies (Huedo-Medina, Sánchez-Meca, Marín-Martínez, & Botella, 2006). Therefore, we can confirm that the random effect model is adequate, and a moderation analysis can be investigated.

Table 5
Meta and weight analysis results for digital management and payment services.

Dependent variable	Independent variable	*O*	*N*	*r*	CI (95%)		Q	I^2	*FNS*	*Egger's Intercep (sig)*	*Weight*	*Type*
Intention to use	Perceived usefulness	13	7069	0.30	0.23	0.36	100.74*	0.88	3527	0.48	0.92	BP
	Trust	9	3550	0.27	0.15	0.39	115.70*	0.93	1161	0.24	0.89	BP
	Perceived ease of use	9	5702	0.26	0.03	0.46	610.74*	0.99	2675	0.61	0.89	BP
	Subjective norms	8	5518	0.25	0.06	0.43	356.55*	0.98	1094	0.82	0.88	BP
	Facilitating conditions	6	2896	0.18	-0.04	0.38	163.73*	0.97	231	0.39	0.67	
	Perceived risk	5	4182	-0.08	-0.16	0.02	29.84*	0.87	39	0.93	0.20	
	Hedonic motivations	4	1486	0.09	-0.15	0.31	59.33*	0.95	5	0.03	0.75	
	Habit	4	2528	0.37	0.28	0.46	19.06*	0.84	588	0.21	1.00	PP
	Attitude	4	868	0.55	0.33	0.71	32.51*	0.94	675	0.81	1.00	PP
	Perceived security	3	869	0.20	-0.07	0.44	32.38*	0.94	49	0.19	0.67	
Intention to continue to use	Satisfaction	3	780	0.65	0.19	0.88	135.67*	0.99	2718	0.13	1.00	PP
	Perceived usefulness	3	1268	0.20	0.10	0.30	6.30*	0.68	55	0.14	1.00	PP
	Trust	3	1176	0.21	0.16	0.26	1.16	0.00	62	0.04	1.00	PP
	Perceived risk	3	1153	-0.15	-0.26	-0.04	7.03*	0.72	30	0.05	0.00	
	Perceived ease of use	3	1637	0.15	0.10	0.21	2.83	0.29	44	0.09	1.00	PP
Attitude	Perceived ease of use	5	1360	0.32	0.22	0.41	14.97*	0.73	323	0.46	1.00	BP
	Perceived usefulness	4	1177	0.40	0.24	0.54	28.90*	0.90	420	0.75	1.00	PP
	Benefits	3	905	0.32	-0.05	0.61	65.48*	0.97	133	0.33	0.67	
Perceived usefulness	Perceived ease of use	5	2257	0.52	0.42	0.60	31.32*	0.87	2118	0.39	1.00	BP

Additionally, to provide an accurate assessment of the asymmetry, the Egger regression resulted to be non-significant, concluding no evidence of publication bias (Naranjo Zolotov et al., 2018).

The results show that out of the 30 relationships, 25 are statistically significant, and five are not statistically significant. Perceived risk to intention to use (r=-0.08), cost to intention to use (r=-0.12), privacy concerns to intention to use (r=0.01), perceived risk to attitude (r=-0.39) and privacy concerns to attitude (r=0.10) are not significant relationships. On the other hand, regarding intention to use digital financial solutions, the top five most significant impacts are from compatibility (r=0.57), attitude (r=0.55), price value (r=0.38), perceived usefulness (r=0.37), and convenience (r=0.36). Regarding intention to continue to use digital financial services, the greatest impacts are satisfaction (r=0.48), trust (r=0.26), perceived usefulness (r=0.25), and perceived ease of use (r=0.14). The most significant impacts on attitude are from perceived usefulness (r=0.39), compatibility (r=0.37), benefits (r=0.36) and subjective norms (r=0.35). Finally, perceived ease of use is statistically significant to perceived usefulness (r=0.55).

After analysing the meta-analysis results, it is relevant to understand the findings of the weight analysis. Weight analysis enables estimating an independent variable's importance, indicating its predictive power to the target variable (Jeyaraj et al., 2006). The weight is calculated by dividing the number of times the independent variable is significant by the total number of relationships between the independent variable and a dependent one. When the weight is 1, all relationships analysed were statistically significant, whereas a weight of 0 indicates no relationship was found to be statistically significant. Moreover, if the variable is tested five or more times, it is "well-utilised". If tested fewer than five times but with a weight equal to 1, the variable is a "promising predictor" (PP). On the other hand, if a variable has a weight higher or equal to 0.8 and was examined five or more times, it is considered a "best predictor" (BP) (Jeyaraj et al., 2006). By combining both a meta and weight analysis, it is possible to get a more comprehensive and complementary view of the significance and strength of the explanatory variables into the target ones. This way, it is possible to identify the strongest predictors.

Of all relationships, 20 were considered best predictors, and two were considered promising predictors. The best predictors of intention to use are perceived usefulness, perceived ease of use, subjective norms, trust, attitude, hedonic motivations, price value, habit, personal innovativeness, convenience, and compatibility. Awareness is considered a promising predictor. Satisfaction, perceived usefulness, trust, and perceived ease of use are best predictors of intention to continue to use. Also, perceived usefulness, perceived ease of use, and subjective norms are best predictors of attitude. Compatibility is a promising indicator. Finally, perceived ease of use is a best predictor of perceived usefulness.

The next step in the analysis was to evaluate the results per digital financial service. The first analysis was conducted for digital banking services, the most common digital financial service. The results are presented in Table 4. Regarding heterogeneity, the results show high heterogeneity levels (around 90%). Regarding the coefficients' significance, the meta-analysis results show that 22 out of 28 relationships are statistically significant. Therefore, the top five variables with the greatest significant impacts on intention to use digital banking are attitude (r=0.58), perceived usefulness (r=0.41), price value (r=0.39), trust (r=0.32), and hedonic motivations (r=0.28). Regarding attitude, the main predictors are perceived usefulness (r=0.39), subjective norms (r=0.39), compatibility (r=0.37), and perceived ease of use (r=0.26). The most significant variables impacting satisfaction are perceived usefulness (r=0.37) and information quality (r=0.21). Regarding intention to continue using digital banking, the significant variables with the greatest impacts are satisfaction (r=0.40), perceived usefulness (r=0.29), and trust (r=0.29). The most influencing factor for digital banking adoption are intention to use (r=0.56), perceived usefulness (r=0.41), and perceived ease of use (0.36). Finally, the results indicate that intention to use positively influences the use behaviour of digital banking (r=0.33 and perceived ease of use impacts perceived usefulness (r=0.58).

Regarding the weight analysis, the best predictors of intention to use digital banking are perceived usefulness, perceived ease of use, attitude, trust, price value, convenience, and personal innovativeness. Perceived usefulness and perceived ease of use are also best predictors of attitude. On the other hand, compatibility and subjective norms are promising indicators of attitude. The promising predictors of satisfaction are information quality and perceived usefulness. Satisfaction is the best predictor of intention to continue to use digital banking, whereas perceived usefulness and trust are promising indicators. Regarding the adoption of digital banking, the best predictor is perceived usefulness, while perceived ease of use and intention to use are promising predictors. Intention to use is a best predictor of use behaviour, and perceived ease of use is a promising indicator of perceived usefulness.

The second analysis was conducted for digital management and payment services only. The results are presented in Table 5. First, the results regarding heterogeneity are very similar to the overall model, showing high heterogeneity levels (around 90%). The results show that five out of 19 relationships are not statistically significant. Therefore, the top five variables with the greatest significant impacts on intention to use digital management and payment services are attitude (r=0.55), habit (r=0.37), perceived usefulness (r=0.30), trust (r=0.27), and

Table 6
Meta and weight analysis results of digital wallet.

Dependent variable	Independent variable	*O*	*N*	*r*	CI (95%)		Q	I^2	*FNS*	*Egger's Intercep (sig)*	*Weight*	*Type*
Intention to use	Perceived usefulness	13	5207	0.38	0.26	0.49	254.11*	0.95	5461	0.55	1.00	BP
	Perceived ease of use	7	1833	0.14	0.08	0.20	10.96*	0.45	91	0.55	1.00	BP
	Trust	7	2132	0.26	0.17	0.34	25.19*	0.76	388	0.34	1.00	BP
	Facilitating conditions	6	1868	0.17	0.03	0.29	39.88*	0.88	103	0.60	0.83	BP
	Perceived risk	4	1236	-6.00	-0.13	0.12	14.85*	0.80	0	0.27	0.50	
	Subjective norms	4	797	0.24	0.17	0.31	1.19	0.00	74	0.97	1.00	PP
	Perceived security	3	1242	0.20	0.11	0.28	3.83	0.48	51	0.12	1.00	PP
	Attitude	3	1348	0.36	0.27	0.44	5.57	0.64	246	0.78	1.00	PP

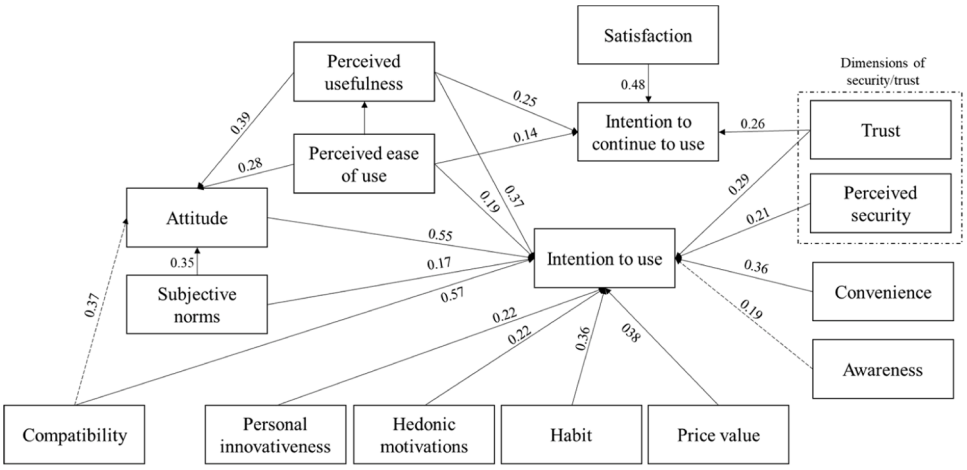


Fig. 4. Model resulting from the meta-analysis of digital financial services
Note: Numerical values represent the average beta. Continuous arrows are "best predictors".

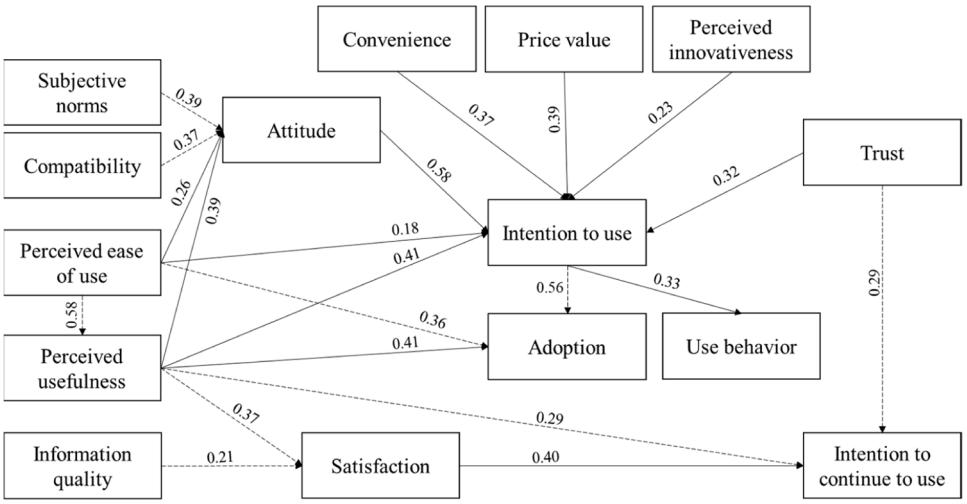


Fig. 5. Model resulting from meta-analysis of digital banking services
Note: Numerical values represent the average beta. Continuous arrows are "best predictors".

perceived ease of use (r=0.26). Regarding intention to continue to use digital management and payment services, the main predictors are satisfaction (r=0.65), trust (r=0.21), perceived usefulness (r=0.20), perceived risk (r=-0.15), and perceived ease of use (r=0.15). The most significant variables impacting attitude are perceived usefulness (r=0.40) and perceived ease of use (r=0.32). Finally, perceived ease of use positively affects perceived usefulness (r=0.52).

Regarding the weight analysis, the best predictors of intention to use digital management and payment services are perceived usefulness, trust, perceived ease of use, and subjective norms. Habit and attitude are promising indicators. Regarding intention to continue to use, satisfaction, perceived usefulness, trust, and perceived ease of use are promising predictors. Perceived ease of use is a best predictor of attitude, while perceived usefulness is a promising predictor. Finally, perceived ease of use is a best predictor of perceived usefulness of digital management and payment services.

Finally, a third analysis was conducted for digital wallets, for which results are presented in Table 6. The results regarding heterogeneity show that three out of the eight relationships did not reject the homogeneity hypothesis. The remaining ones present high heterogeneity

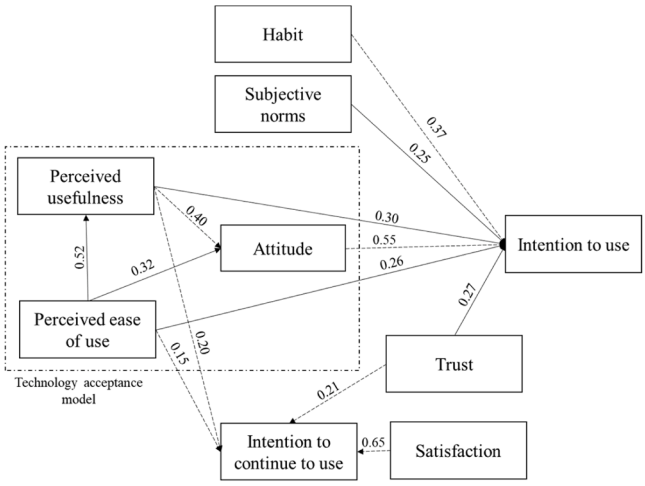


Fig. 6. Model resulting from meta-analysis of digital management and payment services
Note: Numerical values represent the average beta. Continuous arrows are "best predictors".

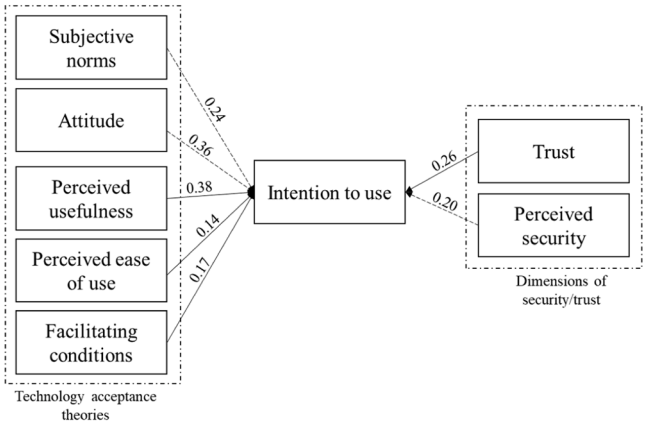


Fig. 7. Model resulting from meta-analysis of digital wallet
Note: Numerical values represent the average beta. Continuous arrows are "best predictors".

Table B.1
Moderation results between perceived usefulness and intention to use digital financial services.

	Estimate	SE	Z	p-value	CI-lb	CI-ub
Intercept	0.41	0.41	0.99	0.32	-0.40	1.21
IDH	0.01	0.49	0.02	0.99	-0.95	0.98
GII	0.01	0.01	1.06	0.29	0.00	0.02
Power distance	0.01	0.00	0.32	0.75	0.00	0.01
Indulgence	0.00	0.00	0.45	0.65	0.00	0.01
Masculinity	0.00	0.00	-1.02	0.31	-0.01	0.00
Uncertainty avoidance	0.00	0.00	-0.52	0.60	-0.01	0.00
Long term orientation	0.00	0.00	-0.40	0.69	-0.01	0.00
Indulgence	-0.01	0.00	-0.41	0.68	-0.01	0.00

levels (around 90%). The results show that one out of eight relationships is not statistically significant. Therefore, the top five variables with the greatest significant impacts on intention to use digital wallets are perceived usefulness (r=0.38), attitude (r=0.36), trust (r=0.26), subjective norms (r=0.24), and perceived security (r=0.20).

Regarding the weight analysis, perceived usefulness, perceived ease of use, trust, and facilitating conditions are best predictors of intention to use a digital wallet. On the other hand, subjective norms, perceived security, and attitude are promising indicators of intention to use.

Table B.2
Moderation results between perceived ease of use and intention to use digital financial services.

	Estimate	SE	Z	p-value	CI-lb	CI-ub
Intercept	-1.99	0.46	-4.35	<0.001	-2.89	-1.09
IDH	1.68	0.48	3.47	0.00	0.74	2.64
GII	0.00	0.01	-0.22	0.82	-0.01	0.01
Power distance	0.01	0.00	3.35	0.00	0.00	0.01
Indulgence	0.00	0.00	-0.36	0.71	-0.01	0.00
Masculinity	0.00	0.01	2.33	0.02	0.00	0.02
Uncertainty avoidance	0.00	0.00	2.20	0.00	0.00	0.01
Long term orientation	0.00	0.00	-1.47	0.14	-0.01	0.00
Indulgence	-0.01	0.00	-2.77	0.01	-0.01	0.00

Table B.3
Moderation results between subjective norms and intention to use digital financial services.

	Estimate	SE	Z	p-value	CI-lb	CI-ub
Intercept	-0.78	0.52	-1.51	0.13	-1.79	0.23
IDH	0.68	0.73	0.93	0.35	-0.75	2.11
GII	-0.01	0.01	-1.63	0.11	-0.03	0.00
Power distance	0.01	0.00	1.72	0.09	0.00	0.01
Indulgence	0.00	0.00	0.66	0.51	-0.01	0.01
Masculinity	0.00	0.01	0.22	0.83	-0.02	0.01
Uncertainty avoidance	0.00	0.00	1.46	0.14	0.00	0.01
Long term orientation	0.01	0.00	1.57	0.12	0.00	0.01
Indulgence	0.00	0.00	0.06	0.95	-0.01	0.01

Table B.4
Moderation results between perceived usefulness and intention to use digital banking.

	Estimate	SE	Z	p-value	CI-lb	CI-ub
Intercept	0.81	0.82	0.99	0.32	-0.79	2.41
IDH	-0.91	0.86	-1.06	0.29	-2.60	0.78
GII	0.01	0.01	1.07	0.29	-0.01	0.04
Power distance	0.00	0.00	0.45	0.65	-0.01	0.01
Indulgence	0.00	0.01	0.66	0.51	-0.01	0.01
Masculinity	0.00	0.01	-0.22	0.83	-0.03	0.02
Uncertainty avoidance	0.00	0.00	-0.69	0.49	-0.01	0.00
Long term orientation	0.00	0.01	-0.40	0.69	-0.01	0.01
Indulgence	0.00	0.00	-0.71	0.48	-0.01	0.00

Moderation results

Along with the meta and weight analyses, a moderation analysis was conducted in order to understand changes in effect size. The results are presented in Appendix B for the relationship between perceived usefulness and intention to use digital financial services (Table B.1), perceived ease of use and intention to use digital financial services (Table B.2), subjective norms and intention to use (Table B.3), and the relationship between perceived usefulness and intention to use digital banking (Table B.4). For this type of analysis, only the relationships that were analysed several times (>20) and present high heterogeneity (Geyskens et al., 2009) can be selected. From all relationships, significant moderators were only found between perceived ease of use and intention to use digital financial services. Therefore, IDH (z=3.47), power distance (z=3.35), masculinity (z=2.33), uncertainty avoidance (z=2.20), and indulgence (z=-2.77) are statistically significant moderators.

Discussion

Digital financial services have been studied over the years, passing

through digital banking, digital management and payment services, not necessarily connected to banks, and the most recent service – digital wallets. The current study indicated that plenty of relationships have been studied. Therefore, from the literature review, 384 relationships were analysed based on 121 quantitative articles. The meta and weight analyses showed the statistically significant and most used constructs to explain a set of target variables.

The results show that the most important predictors of intention to use financial services overall are dimensions from technology acceptance models, such as attitude, perceived usefulness, perceived ease of use, compatibility, and price value. These constructs are widely used in theories like the technology acceptance model (TAM), theory of planned behaviour (TPB), and unified theory of acceptance and use of technology (UTAUT). Nevertheless, previous studies point out that not only technology factors should be considered. Pondering money as a sensitive issue, many articles include security and trust dimensions (Merhi et al., 2019). Especially in these technologies, it is still important to highlight how organisations protect users' data and continually build trust in the service. Trust is also one of the best predictors of intention to continue using digital financial services. Finally, it is also salient to note that individuals who feel more open to innovation or achieve some enjoyment while using these types of solutions are more willing to use them (Rahman et al., 2020), given the great impact of personal innovativeness and hedonic motivations. Fig. 4 presents the meta-analysis results from the overall model of digital financial services, where the continuous arrows represent the best predictors and the non-continuous ones are the promising predictors.

Additionally, moderators were found in the relationship between perceived ease of use and intention to use digital financial technologies, namely on the variables IDH, power distance, masculinity, uncertainty avoidance, and indulgence (with a negative effect). The findings show that the relationship between perceived ease of use and intention to use digital financial services is stronger in countries with a higher human development index. The result is not surprising since cultures with higher income or education usually create individuals more willing to try innovative or digital solutions, like digital financial ones. Moreover, regarding cultural factors, in cultures with high power distance, high masculinity, high uncertainty avoidance and low indulgence, the relationship between perceived ease of use to intention to use is also stronger. Power distance refers to the relationship between a superior and a subordinate (Mora, 2013). Cultures with high power distance tend to accept an inequality distribution of power and accept and not question authority. Also, cultures with higher levels of masculinity tend to clearly have distinct gender roles or unequal pay, for example, where success and money are drivers of society (Hofstede & Minkov, 2010).

Regarding uncertainty avoidance, societies with high uncertainty avoidance tend to make calculated and safe decisions that can lead to expected outcomes, avoiding risky choices (Hofstede & Minkov, 2010). Finally, indulgence refers to the tendency of people in a society to fulfil their desires. Therefore, societies with low indulgence tend to suppress gratification, being more pessimistic and inflexible and characterised by a more pessimistic approach (Hofstede & Minkov, 2010). Overall, the results suggest that more rigid and more money-oriented societies tend to be the ones where the effort to use impacts the use of digital financial services the most. In these cultures, the intention to use these services is higher when the effort to use them is perceived as very low.

Results per main digital financial service

When comparing the three main different digital financial services, the results show that digital banking is the most mature topic, not only because it is the one with more collected articles but also due to the target variables, not only focusing on intention to use, but also on many post-adoption stages (Baptista & Oliveira, 2016). Fig. 5 presents the results of the meta-analysis for digital banking. Overall, the results are very similar to the overall model, showing the relevance of not only technological but also security factors to use digital banking.

Regarding digital management and payment services, the main difference from the other services is the presence of habit as a best predictor of intention to use, as seen in Fig. 6. Habit is defined as the extent to which individuals perform behaviours due to learning or routine (Venkatesh et al., 2012). This finding suggests that people with a certain familiarity with this service will present greater intention to use them. In fact, this type of digital service tends to require more knowledge since they are frequently used to invest money and define investment positions (Gerlach & Lutz, 2021). Therefore, only people that feel more comfortable with these services will be willing to use them.

On the other hand, the least mature topic is digital wallets, as can be seen in Fig. 7. The Meta-analysis results are somewhat similar to the overall model, showing both technological and security dimensions as primary motivators to use. However, it is relevant to notice the facilitating conditions factor is not present in the other two models. Facilitating conditions measure the degree to which individuals believe they have the necessary resources to use a technology or that organisational and technical support exists when needed (Venkatesh et al., 2012). This finding suggests that individuals are more willing to use mobile wallets when feeling that support for their use exists, both from an infrastructural point of view or the provider organisations, proving the relevant need of companies to pass a supportive and trustworthy image to their customers (Jaiswal et al., 2022).

Theoretical implications

Concerning implications for theory, this work provides a comprehensive picture of digital financial services, overall and per main service area. The meta-analysis allowed to have a cumulated influence of each explanatory variable on the dependent variable, as well as an evaluation of the significance, while the weight analysis provided a weight. Therefore, it was possible to obtain a model of the best predictors of digital financial services use. Based on this, we conclude that the main factors used are the dimensions from technological acceptance theories, like TAM, TPB and UTAUT – even on more recent investigations (e.g. Alnemer, 2022; Khan, 2022) -, adding dimensions related to trust and security, indicating that other constructs related to these security dimensions should be continually tested, as they are proven to be strong predictors, as found in previous relevant works (Oliveira et al., 2016; Talwar et al., 2020). Therefore, the resulting models provide both a synthesis and a support basis for future research. Moreover, it contributes to a better understanding of the tendencies on the digital financial services field, proving the relevance of new services, like digital wallets (Chawla & Joshi, 2021), that can still further be investigated, since the number of used predictors is low, compared to the other services that are more mature. This shows that there is still room to deep investigate these digital wallets (Mohd Thas Thaker et al., 2022), where the provided synthesis model can be a starting point. For example, recent research is starting to analyse how digital wallets can affect the financial stability and awareness of the user, or if it impulses more purchases (Lee et al., 2022). Additionally, from the weight analysis, we can see that although variables like facilitating conditions, perceived risk, and benefits have been used continuously, their weight is low, meaning that they have been found as continuously not significant. From these results, researchers can make use of these findings to help in the selection of factors to analyse the use of digital financial services. For example, promising indicators can be further explored, while the non-significant and with low weight variables can be disregarded. Overall, the same theories have been applied to different digital financial services, however, other theories could also be tested, making use of the socio-technical perspective of IS, for example, to study antecedents of use, but also extending to it to the study of outcomes (Sarker et al., 2019), as well as understanding the use of different intelligent systems, like Internet of Things, virtual reality (Arjun et al., 2021), cloud-based blockchain to also improve financial security (Amponsah et al., 2022).

For example, understand how the inclusion of artificial intelligence can help in performing financial investments through these digital services (Akbarighatar et al., 2023).

Practical implications

Regarding practical implications, the results provide relevant suggestions to the practitioner side in order to increase the use of digital financial services. The results suggest that besides the standard technological dimensions, it is important that organisations show how data is protected and treated, creating a trustworthy environment. The security dimensions play a major role in these digital services; as shown, they are present in all types of digital financial services. Another important finding is that individuals who are more familiar or have more support to use digital financial services are more willing to use them. Therefore, it is important to engage users through knowledgeable interaction, like demonstrations, and overall increase the digital literacy of individuals.

Limitations and future research

The study is not without limitations. The first one is related to the articles used to perform the analysis. Although 121 articles are considered a significant number of articles, some of the existing studies had to be discarded due to the inexistence of correlation coefficients and/or sample sizes. This way, only quantitative articles were selected. Moreover, when analysing digital wallets per se, only 18 articles were available; however, we find it reasonable given the topic's novelty. Also, most articles analysed only one country, for which we recommend future studies with more country comparisons, given the strong prospect of cultural values to influence technological behaviours. Finally, most studies have focused on technological and security dimensions, which leaves the question if other theories or constructs could also be relevant. For example, given the sensitiveness of data, theories that investigate privacy issues, like privacy calculus theory, or the protection motivation theory, that comprises the individuals' perceptions about the risk's preventive behaviours. Further studies could analyse and/or validate this, resorting to a mixed methods design, for example, to better understand consumer behaviour regarding digital financial services.

Conclusion

The study of digital financial services started many years ago. However, digital financial solutions have gained renewed attention, primarily as preventive COVID-19 solutions have been implemented to avoid face-to-face interactions and facilitate the performance of tasks anywhere and anytime. Therefore, given the vast research on the topic, we applied a weight and meta-analysis based on 121 articles, resulting in 384 identified relationships. The study was conducted for all digital financial services, but also individually for the three main identified areas: digital banking – the most common, digital management and payment services, and digital wallets – the most recent service. As a result, this study presents a comprehensive view of the best and most promising predictors of the behavioural intention to use digital financial services, emphasising the importance of technological, but also security and trust dimensions for individuals to use these solutions.

Moreover, the moderation analysis results show that more money-oriented cultures tend to use these services more when perceiving them as easy to use. Overall, this study not only illustrates the state of the art of digital financial services but also serves as support for future research, opening the door to new theories that might affect users' behaviours that can go beyond technological and security factors. The study also strongly supports the practitioner side, providing several recommendations for digital financial services use strategies.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Variables codification

Final construct	Original constructs
Age	Age
Anxiety	Anxiety
Assurance	Assurance, Security assurance
Attitude	Attitude, Attitude toward behaviour
Autonomy	Autonomy, Perceived autonomy
Awareness	Awareness, Brand awareness
Benefits	Benefits, Economic benefit, Perceived benefitinformation security benefit perceived benefit price benefit promotional benefit
Benevolence	Benevolence
Coercive pressure	Coercive pressure
Communication	Communication
Compatibility	Compatibility, Perceived compatibility
Competence	Competence, Perceived competence, Technology competency
Complexity	Complexity
Confirmation	Confirmation
Convenience	Convenience, Benefit convenience, Convenience of system
Cost	Cost, Perceived cost
Credibility	Credibility
Customer support	Customer support
Design	Design, Design aesthetic, Design quality
Digital literacy	Digital literacy
Discomfort	Discomfort

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Final construct	Original constructs
Education	Education
Efficiency	Efficiency
Empowerment	Psychological empowerment
Engagement	Engagement
Experience	Brand experience, Experience
External regulation	External regulation
External risk	External risk
Facilitating conditions	Facilitating conditions
Financial risk	Financial risk
Functional failure	Functional failure
Gamification impact	Gamification impact
Government support	Government support
Habit	Habit
Hedonic motivations	Enjoyment, Hedonic motivations, Perceived enjoyment
Image	Brand image, Image
Income	Income
Information access	Information access
Information quality	Information quality
Intention to use	Behavioural intention, Intention to use, Usage intention
Internal risk	Internal risk
Intrinsic motivation	Intrinsic motivation
Investment diversity	Investment diversity
Knowledge	Digitisation knowledge, Knowledge
Lack of support	Lack of support
Observability	Observability
Final construct	**Original constructs**
Optimism	Optimism
Perceived behavioural control	Perceived behavioural control
Perceived covid 19 risk	Perceived covid 19 health risk, Perceived covid 19 risk
Perceived ease of use	Ease of use, Effort expectancy, Perceived ease of use
Perceived gamification	Perceived gamification
Perceived quality	Perceived quality
Perceived risk	Perceived risk, Perception of risk, Risk
Perceived security	Security, Perceived security
Perceived usefulness	Perceived usefulness, Performance expectancy, Relative advantage
Performance risk	Performance risk
Personal innovativeness	Consumer innovativeness, Innovativeness, Personal innovativeness
Price value	Price, Price value
Privacy concern	Privacy concern, Privacy risk
Regulatory support	Regulatory support
Reliability	Reliability
Resistance	Resistance
Responsiveness	Responsiveness
Satisfaction	Satisfaction, Use satisfaction
Self-efficacy	Self-efficacy
Service quality	Service quality
Subjective norms	Social influence, Social norms, Subjective norm
System quality	System quality
Tangibility	Tangibility
Task characteristics	Task characteristics
Task technology fit	Task technology fit
Technical risk	Technical risk
Technology characteristics	Technology characteristics
Technology readiness	Technology readiness
Technology support	Technology support
Time risk	Time risk
Trialability	Trialability
Trust	Bank trust, Initial trust, Trust, Perceived trust
Uncertainty avoidance	Uncertainty avoidance
Use barrier	Usage barrier
Word of mouth	Word of mouth
Adoption	Adoption
Continuous use	Continued usage
Individual performance	Individual performance
Intention to continue to use	Continuance intention, Intention to continue to use
Intention to recommend	Intention to recommend, Recommendation intention

Appendix B. Moderation results

References

Abbasi, G. A., Sandran, T., Ganesan, Y., & Iranmanesh, M. (2022). Go cashless! Determinants of continuance intention to use E-wallet apps: A hybrid approach using PLS-SEM and fsQCA. *Technology in Society, 68*. https://doi.org/10.1016/j.techsoc.2022.101937

Abegao Neto, & Figueiredo, J. C. B. D. (2022). Effects of age and income moderation on adoption of mobile payments in Brazil. *Innovation & Management Review*. https://doi.org/10.1108/INMR-06-2021-0109

Abikari, M., Öhman, P., & Yazdanfar, D. (2022). Negative emotions and consumer behavioural intention to adopt emerging e-banking technology. *Journal of Financial Services Marketing*. https://doi.org/10.1057/s41264-022-00172-x

Aboobucker, I., & Bao, Y. (2018). What obstruct customer acceptance of internet banking? Security and privacy, risk, trust and website usability and the role of moderators. *Journal of High Technology Management Research, 29*(1), 109–123. https://doi.org/10.1016/j.hitech.2018.04.010

Adjei, J. K., Odei-Appiah, S., & Tobbin, P. E. (2020). Explaining the determinants of continual use of mobile financial services. *Digital Policy, Regulation and Governance, 22*(1), 15–31. https://doi.org/10.1108/DPRG-09-2019-0074

Afshan, S., & Sharif, A. (2016). Acceptance of mobile banking framework in Pakistan. *Telematics and Informatics, 33*(2), 370–387. https://doi.org/10.1016/j.tele.2015.09.005

Agárdi, I., & Alt, M. A. M. A. (2022). Do digital natives use mobile payment differently than digital immigrants? A comparative study between generation X and Z. *Electronic Commerce Research*. https://doi.org/10.1007/s10660-022-09537-9

Agyei, J., Sun, S., Penney, E. K., Abrokwah, E., Boadi, E. K., & Fiifi, D. (2022). Internet banking services user adoption in Ghana: An empirical study. *Journal of African Business, 23*(3), 599–616. https://doi.org/10.1080/15228916.2021.1904756

Ahmad, S., Bhatti, S. H., & Hwang, Y. (2020). E-service quality and actual use of e-banking: Explanation through the technology acceptance model. *Information Development, 36*(4), 503–519. https://doi.org/10.1177/0266666919871611

Aji, H. M., Berakon, I., & Md Husin, M. (2020). COVID-19 and e-wallet usage intention: A multigroup analysis between Indonesia and Malaysia. *Cogent Business and Management, 7*(1). https://doi.org/10.1080/23311975.2020.1804181

Akbarighatar, P., Pappas, I., & Vassilakopoulou, P. (2023). A sociotechnical perspective for responsible AI maturity models: Findings from a mixed-method literature review. *International Journal of Information Management Data Insights, 3*, Article 100193. https://doi.org/10.1016/j.jjimei.2023.100193

Akhter, A., Karim, M. M. M. M., Jannat, S., & Anwarul Islam, K. M. M. (2022). Determining factors of intention to adopt internet banking services: A study on commercial bank users in Bangladesh. *Banks and Bank Systems, 17*(1), 125–136. https://doi.org/10.21511/BBS.17(1).2022.11

Akturan, U., & Tezcan, N. (2012). Mobile banking adoption of the youth market: Perceptions and intentions. *Marketing Intelligence and Planning, 30*(4), 444–459. https://doi.org/10.1108/02634501211231928

Al-Ajam, A. S., & Md Nor, K. (2015). Challenges of adoption of internet banking service in yemen. *International Journal of Bank Marketing, 33*(2), 178–194. https://doi.org/10.1108/IJBM-01-2013-0001

Al-Dmour, H. H., Ali, W. K., & Al-Dmour, R. H. (2019). The relationship between customer engagement, satisfaction, and loyalty. *International Journal of Customer Relationship Marketing and Management, 10*(2), 35–60. https://doi.org/10.4018/IJCRMM.2019040103

Al-Hattami, H. M., Abdullah, A. A. H., & Khamis, A. A. A. (2021). Determinants of intention to continue using internet banking: Indian context. *Innovative Marketing, 17*(1), 40–52. https://doi.org/10.21511/im.17(1).2021.04

Al-Okaily, M., Alkhwaldi, A. F. A. F., Abdulmuhsin, A. A. A. A., Alqudah, H., & Al-Okaily, A. (2022). Cloud-based accounting information systems usage and its impact on Jordanian SMEs' performance: The post-COVID-19 perspective. *Journal of Financial Reporting and Accounting*. https://doi.org/10.1108/JFRA-12-2021-0476

Al-Okaily, M., Lutfi, A., Alsaad, A., Taamneh, A., & Alsyouf, A. (2020). The determinants of digital payment systems' acceptance under cultural orientation differences: The case of uncertainty avoidance. *Technology in Society, 63*, Article 101367. https://doi.org/10.1016/j.techsoc.2020.101367

Al-Qudah, A. A., Al-Okaily, M., Alqudah, G., & Ghazlat, A. (2022). Mobile payment adoption in the time of the COVID-19 pandemic. *Electronic Commerce Research*. https://doi.org/10.1007/s10660-022-09577-1

Al-Sabaawi, M. Y. M., Alshaher, A. A., & Alsalem, M. A. (2021). User trends of electronic payment systems adoption in developing countries: An empirical analysis. *Journal of Science and Technology Policy Management*. https://doi.org/10.1108/JSTPM-11-2020-0162

Al-Zu'bi, K., & Al-Gasawneh, J. A. J. A (2022). An integrated model of mobile banking service quality and customers' satisfaction: Evidence from Jordanian mobile banking users. *International Journal of Data and Network Science, 6*(4), 1609–1618. https://doi.org/10.5267/j.ijdns.2022.4.017

Al Khasawneh, M. H. (2015). An empirical examination of consumer adoption of mobile banking (M-banking) in Jordan. *Journal of Internet Commerce, 14*(3), 341–362. https://doi.org/10.1080/15332861.2015.1045288

Alalwan, A. A., Dwivedi, Y. K., & Rana, N. P. (2017). Factors influencing adoption of mobile banking by Jordanian bank customers: Extending UTAUT2 with trust. *International Journal of Information Management, 37*(3), 99–110. https://doi.org/10.1016/j.ijinfomgt.2017.01.002

Almaiah, M. A. M. A., Al-Rahmi, A. M. A. M., Alturise, F., Alrawad, M., Alkhalaf, S., Lutfi, A., Al-Rahmi, W. M. W. M., & Awad, A. B. A. B. (2022). Factors influencing the adoption of internet banking: An integration of ISSM and UTAUT with price value and perceived risk. *Frontiers in Psychology, 13*. https://doi.org/10.3389/fpsyg.2022.919198

Alnemer, H. A. (2022). Determinants of digital banking adoption in the Kingdom of Saudi Arabia: A technology acceptance model approach. *Digital Business, 2*(2), Article 100037. https://doi.org/10.1016/j.digbus.2022.100037

Alshurideh, M. T., Al Kurdi, B., Masa'deh, R., & Salloum, S. A (2021). The moderation effect of gender on accepting electronic payment technology: A study on United Arab Emirates consumers. *Review of International Business and Strategy, 31*(3), 375–396. https://doi.org/10.1108/RIBS-08-2020-0102

Amponsah, A. A., Adekoya, A. F., & Weyori, B. A. (2022). Improving the financial security of National Health Insurance using cloud-based blockchain technology application. *International Journal of Information Management Data Insights, 2*(1), Article 100081. https://doi.org/10.1016/j.jjimei.2022.100081

Anouze, A. L. M., & Alamro, A. S. (2020). Factors affecting intention to use e-banking in Jordan. *International Journal of Bank Marketing, 38*(1), 86–112. https://doi.org/10.1108/IJBM-10-2018-0271

Arjun, R., Kuanr, A., & Kr, S. (2021). Developing banking intelligence in emerging markets : Systematic review and agenda. *International Journal of Information Management Data Insights, 1*(2), Article 100026. https://doi.org/10.1016/j.jjimei.2021.100026

Baabdullah, A. M., Alalwan, A. A., Rana, N. P., Patil, P., & Dwivedi, Y. K. (2019). An integrated model for m-banking adoption in Saudi Arabia. *International Journal of Bank Marketing, 37*(2), 452–478. https://doi.org/10.1108/IJBM-07-2018-0183

Astari, A., Yasa, N., Sukaatmadja, I., & Giantari, I. G. A. K. (2022). Integration of technology acceptance model (TAM) and theory of planned behavior (TPB): An e-wallet behavior with fear of COVID-19 as a moderator variable. *International Journal of Data and Network Science, 6*(4), 1427–1436. https://doi.org/10.5267/j.ijdns.2022.5.008

Bailey, A. A. A., Bonifield, C. M. C. M., Arias, A., & Villegas, J. (2022). Mobile payment adoption in Latin America. *Journal of Services Marketing, 36*(8), 1058–1075. https://doi.org/10.1108/JSM-04-2021-0130

Banerjee, S., & Sreejesh, S. (2022). Examining the role of customers' intrinsic motivation on continued usage of mobile banking: A relational approach. *International Journal of Bank Marketing, 40*(1), 87–109. https://doi.org/10.1108/IJBM-06-2021-0216

Bankuoru Egala, S., Boateng, D., & Aboagye Mensah, S. (2021). To leave or retain? An interplay between quality digital banking services and customer satisfaction. *International Journal of Bank Marketing, 39*(7), 1420–1445. https://doi.org/10.1108/IJBM-02-2021-0072

Baptista, G., & Oliveira, T. (2017). Why so serious? Gamification impact in the acceptance of mobile banking services. *Internet Research, 27*(1), 118–139. https://doi.org/10.1108/IntR-10-2015-0295

Baptista, G., & Oliveira, T. (2015). Understanding mobile banking: The unified theory of acceptance and use of technology combined with cultural moderators. *Computers in Human Behavior, 50*, 418–430. https://doi.org/10.1016/J.CHB.2015.04.024

Baptista, G., & Oliveira, T. (2016). A weight and a meta-analysis on mobile banking acceptance research. *Computers in Human Behavior, 63*, 480–489. https://doi.org/10.1016/j.chb.2016.05.074

Bitrián, P., Buil, I., & Catalán, S. (2021). Making finance fun: The gamification of personal financial management apps. *International Journal of Bank Marketing, 39*(7), 1310–1332. https://doi.org/10.1108/IJBM-02-2021-0074

Borenstein, M., Hedges, L. V., Higgins, J. P. T., & Rothstein, H. R. (2010). A basic introduction to fixed-effect and random-effects models for meta-analysis. *Research Synthesis Methods, 1*(2), 97–111. https://doi.org/10.1002/jrsm.12

Borenstein, M. L., Hedges, V., L, P. T., Higgins, J., & Rothstein, H (2009). Criticisms of meta-analysis. *Introduction to meta-analysis* (p. Chapter 43).

Changchit, C., Lonkani, R., & Sampet, J. (2017). Mobile banking: Exploring determinants of its adoption. *Journal of Organizational Computing and Electronic Commerce, 27*(3), 239–261. https://doi.org/10.1080/10919392.2017.1332145

Chaouali, W., Ben Yahia, I., Lunardo, R., & Triki, A. (2019). Reconsidering the "what is beautiful is good" effect: When and how design aesthetics affect intentions towards mobile banking applications. *International Journal of Bank Marketing, 37*(7), 1525–1546. https://doi.org/10.1108/IJBM-12-2018-0337

Chaouali, W., & El Hedhli, K. (2019). Toward a contagion-based model of mobile banking adoption. *International Journal of Bank Marketing, 37*(1), 69–96. https://doi.org/10.1108/IJBM-05-2017-0096

Chauhan, V., Yadav, R., & Choudhary, V. (2019). Analyzing the impact of consumer innovativeness and perceived risk in internet banking adoption: A study of Indian consumers. *International Journal of Bank Marketing, 37*(1), 323–339. https://doi.org/10.1108/IJBM-02-2018-0028

Chawla, D., & Joshi, H. (2019). Consumer attitude and intention to adopt mobile wallet in India – An empirical study. *International Journal of Bank Marketing, 37*(7), 1590–1618. https://doi.org/10.1108/IJBM-09-2018-0256

Chawla, D., & Joshi, H. (2021). Degree of awareness and the antecedents of the digital media platform: The case of mobile wallets. *FIIB Business Review*. https://doi.org/10.1177/23197145211023413

Chemingui, H., & Lallouna, H. B. (2013). Resistance, motivations, trust and intention to use mobile financial services. *International Journal of Bank Marketing, 31*(7), 574–592. https://doi.org/10.1108/IJBM-12-2012-0124

Chen, C. S. (2013). Perceived risk, usage frequency of mobile banking services. *Managing Service Quality, 23*(5), 410–436. https://doi.org/10.1108/MSQ-10-2012-0137

Chen, X., & Li, S. (2017). Understanding continuance intention of mobile payment services: An empirical study. *Journal of Computer Information Systems, 57*(4), 287–298. https://doi.org/10.1080/08874417.2016.1180649

Chong, A. Y.-L., Ooi, K.-B., Lin, B., & Tan, B.-I. (2010). Online banking adoption: An empirical analysis. *International Journal of Bank Marketing, 28*(4), 267–287. https://doi.org/10.1108/02652321011054963

Ciunova-Shuleska, A., Palamidovska-Sterjadovska, N., & Prodanova, J. (2022). What drives m-banking clients to continue using m-banking services? *Journal of Business Research, 139*, 731–739. https://doi.org/10.1016/j.jbusres.2021.10.024

Cook, T. D. (1991). Meta-analysis: Its potential for causal description and causal explanation within program evaluation. *Social prevention and the social sciences* (pp. 245–286). De Gruyter. https://doi.org/10.1515/9783110864328.245

Deb, M., & Lomo-David, E. (2014). An empirical examination of customers' adoption of m-banking in India. *Marketing Intelligence and Planning, 32*(4), 475–494. https://doi.org/10.1108/MIP-07-2013-0119

Egger, M., Smith, G. D., Schneider, M., & Minder, C. (1997). Bias in meta-analysis detected by a simple, graphical test. *British Medical Journal, 315*(7109), 629–634. https://doi.org/10.1136/bmj.315.7109.629

Elhajjar, S., & Ouaida, F. (2020). An analysis of factors affecting mobile banking adoption. *International Journal of Bank Marketing, 38*(2), 352–367. https://doi.org/10.1108/IJBM-02-2019-0055

Farah, M. F., Hasni, M. J. S., & Abbas, A. K. (2018). Mobile-banking adoption: Empirical evidence from the banking sector in Pakistan. *International Journal of Bank Marketing, 36*(7), 1386–1413. https://doi.org/10.1108/IJBM-10-2017-0215

Franque, F. B., Oliveira, T., Tam, C., & Santini, F.deO. (2021). A meta-analysis of the quantitative studies in continuance intention to use an information system. *Internet Research, 31*(1), 123–158. https://doi.org/10.1108/INTR-03-2019-0103/FULL/PDF

Garzaro, D. M., Varotto, L. F., & Pedro, S. C. (2021). Internet and mobile banking: The role of engagement and experience on satisfaction and loyalty. *International Journal of Bank Marketing, 39*(1), 1–23. https://doi.org/10.1108/IJBM-08-2020-0457

George, A., & Sunny, P. (2022). Why do people continue using mobile wallets? An empirical analysis amid COVID-19 pandemic. *Journal of Financial Services Marketing*. https://doi.org/10.1057/s41264-022-00174-9

Gerlach, J. M., & Lutz, J. K. T. (2021). Digital financial advice solutions – Evidence on factors affecting the future usage intention and the moderating effect of experience. *Journal of Economics and Business, 117*. https://doi.org/10.1016/j.jeconbus.2021.106009

Geyskens, I., Krishnan, R., Steenkamp, J. B. E. M., & Cunha, P. V. (2009). A review and evaluation of meta-analysis practices in management research. *Journal of Management, 35*(2), 393–419. https://doi.org/10.1177/0149206308328501

Glavee-Geo, R., Shaikh, A. A., & Karjaluoto, H. (2017). Mobile banking services adoption in Pakistan: Are there gender differences? *International Journal of Bank Marketing, 35*(7), 1088–1112. https://doi.org/10.1108/IJBM-09-2015-0142

Gupta, A., & Arora, N. (2017). Consumer adoption of m-banking: A behavioral reasoning theory perspective. *International Journal of Bank Marketing, 35*(4), 733–747. https://doi.org/10.1108/IJBM-11-2016-0162

Gupta, K., & Arora, N. (2020). Investigating consumer intention to accept mobile payment systems through unified theory of acceptance model: An Indian perspective. *South Asian Journal of Business Studies, 9*(1), 88–114. https://doi.org/10.1108/SAJBS-03-2019-0037

Gupta, S., Dhingra, S., Tanwar, S., & Aggarwal, R. (2022). What explains the adoption of mobile wallets? A study from merchants' perspectives. *International Journal of Human-Computer Interaction*. https://doi.org/10.1080/10447318.2022.2104408

Hanafizadeh, P., Behboudi, M., Abedini Koshksaray, A., & Jalilvand Shirkhani Tabar, M. (2014). Mobile-banking adoption by Iranian bank clients. *Telematics and Informatics, 31*(1), 62–78. https://doi.org/10.1016/j.tele.2012.11.001

Hassan, H. E., & Wood, V. R. (2020). Does country culture influence consumers' perceptions toward mobile banking? A comparison between Egypt and the United States. *Telematics and Informatics, 46*, Article 101312. https://doi.org/10.1016/j.tele.2019.101312

Hidayat-Ur-Rehman, I., Ahmad, A., Khan, M. N., & Mokhtar, S. A. (2021). Investigating mobile banking continuance intention: A mixed-methods approach. *Mobile Information Systems, 2021*. https://doi.org/10.1155/2021/9994990

Higgins, J. P. T., & Thompson, S. G. (2002). Quantifying heterogeneity in a meta-analysis. *Statistics in Medicine, 21*(11), 1539–1558. https://doi.org/10.1002/sim.1186

Ho, J. C., Wu, C. G., Lee, C. S., & Pham, T. T. T. (2020). Factors affecting the behavioral intention to adopt mobile banking: An international comparison. *Technology in Society, 63*, 101360. 10.1016/j.techsoc.2020.101360.

Hofstede, G., & Minkov, M. (2010). *Cultures and Organizations: Software of the Mind* (Third Edition). Espanha: McGraw Hill LLC.

Huedo-Medina, T. B., Sánchez-Meca, J., Marín-Martínez, F., & Botella, J. (2006). Assessing heterogeneity in meta-analysis: Q statistic or I2 index? *Psychological methods, 11*(2), 193. https://doi.org/10.1037/1082-989X.11.2.193

Humbani, M., & Wiese, M. (2019). An integrated framework for the adoption and continuance intention to use mobile payment apps. *International Journal of Bank Marketing, 37*(2), 646–664. https://doi.org/10.1108/IJBM-03-2018-0072

Ibrahim, M., Hinson, R. E. R. E., & Stephen, A. (2022). Exploring consumers' intention to adopt mobile payment systems in Ghana. *International Journal of E-Services and Mobile Applications, 14*(1). https://doi.org/10.4018/IJESMA.285547

Idoga, P. E., Toycan, M., Nadiri, H., & Çelebi, E. (2018). Factors affecting the successful adoption of e-health cloud based health system from healthcare consumers' perspective. *IEEE Access, 6*, 71216–71228. https://doi.org/10.1109/ACCESS.2018.2881489

Inan, D. I., Hidayanto, A. N., Juita, R., Soemawilaga, F. F., Melinda, F., Puspacinantya, P., & Amalia, Y. (2021). Service quality and self-determination theory towards continuance usage intention of mobile banking. *Journal of Science and Technology Policy Management*. https://doi.org/10.1108/JSTPM-01-2021-0005

Inder, S., Sood, K., & Grima, S. (2022). Antecedents of behavioural intention to adopt internet banking using structural equation modelling. *Journal of Risk and Financial Management, 15*(4). https://doi.org/10.3390/jrfm15040157

Jaiswal, D., Kaushal, V., Mohan, A., & Thaichon, P. (2022). Mobile wallets adoption: Pre- and post-adoption dynamics of mobile wallets usage. *Marketing Intelligence and Planning, 40*(5), 573–588. https://doi.org/10.1108/MIP-12-2021-0466

Jebarajakirthy, C., & Shankar, A. (2021). Impact of online convenience on mobile banking adoption intention: A moderated mediation approach. *Journal of Retailing and Consumer Services, 58*. https://doi.org/10.1016/j.jretconser.2020.102323

Jena, R. K. R. K. (2022). Investigating and predicting intentions to continue using mobile payment platforms after the COVID-19 pandemic: An empirical study among retailers in India. *Journal of Risk and Financial Management, 15*(7). https://doi.org/10.3390/jrfm15070314

Jeyaraj, A., Rottman, J. W., & Lacity, M. C. (2006). A review of the predictors, linkages, and biases in IT innovation adoption research. *Journal of information technology (Vol. 21, issue 1, pp. 1–23)*. London, England: SAGE PublicationsSage UK. https://doi.org/10.1057/palgrave.jit.2000056

Kamboj, S., Sharma, M., & Sarmah, B. (2022). Impact of mobile banking failure on bank customers' usage behaviour: The mediating role of user satisfaction. *International Journal of Bank Marketing, 40*(1), 128–153. https://doi.org/10.1108/IJBM-10-2020-0534

Kapoor, A., Sindwani, R., Goel, M., & Shankar, A. (2022). Mobile wallet adoption intention amid COVID-19 pandemic outbreak: A novel conceptual framework. *Computers and Industrial Engineering, 172*, Article 108646. https://doi.org/10.1016/j.cie.2022.108646

Kaur, P., Dhir, A., Bodhi, R., Singh, T., & Almotairi, M. (2020). Why do people use and recommend m-wallets? *Journal of Retailing and Consumer Services, 56*. https://doi.org/10.1016/j.jretconser.2020.102091

Kaur, S., & Arora, S. (2021). Role of perceived risk in online banking and its impact on behavioral intention: Trust as a moderator. *Journal of Asia Business Studies, 15*(1), 1–30. https://doi.org/10.1108/JABS-08-2019-0252

Kesharwani, A., & Bisht, S. S. (2012). The impact of trust and perceived risk on internet banking adoption in India: An extension of technology acceptance model. *International Journal of Bank Marketing, 30*(4), 303–322. https://doi.org/10.1108/02652321211236923

Khan, A. G., Lima, R. P., & Mahmud, M. S. (2021). Understanding the service quality and customer satisfaction of mobile banking in Bangladesh: Using a structural equation model. *Global Business Review, 22*(1), 85–100. https://doi.org/10.1177/0972150918795551

Khan, I. U. (2022). How does culture influence digital banking? A comparative study based on the unified model. *Technology in Society, 68*, Article 101822. https://doi.org/10.1016/j.techsoc.2021.101822

Khoa, B. T. (2020). The impact of the personal data disclosure's tradeoff on the trust and attitude loyalty in mobile banking services. *Journal of Promotion Management, 27*(4), 585–608. https://doi.org/10.1080/10496491.2020.1838028

Kim, C., Mirusmonov, M., & Lee, I. (2010). An empirical examination of factors influencing the intention to use mobile payment. *Computers in Human Behavior, 26*(3), 310–322. https://doi.org/10.1016/j.chb.2009.10.013

Koenig-Lewis, N., Marquet, M., Palmer, A., & Zhao, A. L. (2015). Enjoyment and social influence: Predicting mobile payment adoption. *Service Industries Journal, 35*(10), 537–554. https://doi.org/10.1080/02642069.2015.1043278

Kumar, A., Dhingra, S., Batra, V., & Purohit, H. (2020). A framework of mobile banking adoption in India. *Journal of Open Innovation: Technology, Market, and Complexity, 6*(2). https://doi.org/10.3390/JOITMC6020040

Lee, Y. Y., Gan, C. L., & Liew, T. W. (2022). Do E-wallets trigger impulse purchases? An analysis of Malaysian Gen-Y and Gen-Z consumers. *Journal of Marketing Analytics*, 244–261. https://doi.org/10.1057/s41270-022-00164-9

Leong, L. Y., Hew, T. S., Ooi, K. B., & Wei, J. (2020). Predicting mobile wallet resistance: A two-staged structural equation modeling-artificial neural network approach. *International Journal of Information Management, 51*, Article 102047. https://doi.org/10.1016/j.ijinfomgt.2019.102047

Lian, J. W., & Li, J. (2021). The dimensions of trust:An investigation of mobile payment services in Taiwan. *Technology in Society, 67*. https://doi.org/10.1016/j.techsoc.2021.101753

Lifen Zhao, A., Koenig-Lewis, N., Hanmer-Lloyd, S., & Ward, P (2010). Adoption of internet banking services in China: Is it all about trust? *International Journal of Bank Marketing, 28*(1), 7–26. https://doi.org/10.1108/02652321011013562

Lin, H.-F. (2011). An empirical investigation of mobile banking adoption: The effect of innovation attributes and knowledge-based trust. *International Journal of Information Management, 31*(3), 252–260. https://doi.org/10.1016/j.ijinfomgt.2010.07.006

Lisana, & Handarkho, Y. D. Y. D (2022). Social aspect versus service quality in trust formation toward mobile payment adoption: A case study of Indonesia. *Asia Pacific Journal of Marketing and Logistics*. https://doi.org/10.1108/APJML-10-2021-0774

Liu, Y., Wang, M., Huang, D., Huang, Q., Yang, H., & Li, Z. (2019). The impact of mobility, risk, and cost on the users' intention to adopt mobile payments. *Information Systems and E-Business Management, 17*(2–4), 319–342. https://doi.org/10.1007/s10257-019-00449-0

Ly, B., & Ly, R. (2022). Internet banking adoption under technology acceptance model—Evidence from Cambodian users. *Computers in Human Behavior Reports, 7*, Article 100224. https://doi.org/10.1016/j.chbr.2022.100224

Ly, H. T. N., Khuong, N. V., & Son, T. H. (2022). Determinants affect mobile wallet continuous usage in Covid 19 pandemic: Evidence from Vietnam. *Cogent Business and Management, 9*(1). https://doi.org/10.1080/23311975.2022.2041792

Madan, K., & Yadav, R. (2016). Behavioural intention to adopt mobile wallet: A developing country perspective. *Journal of Indian Business Research, 8*(3), 227–244. https://doi.org/10.1108/JIBR-10-2015-0112

Maher, M., Khan, I., & Prikshat, V. (2023). Monetisation of digital health data through a GDPR-compliant and blockchain enabled digital health data marketplace : A proposal to enhance patient ' s engagement with health data repositories.

International Journal of Information Management Data Insights, 3(1), Article 100159. https://doi.org/10.1016/j.jjimei.2023.100159
Mainardes, E. W., Costa, P. M. F., & Nossa, S. N. (2023). Customers' satisfaction with fintech services: Evidence from Brazil. *Journal of Financial Services Marketing, 28*(2), 378–395. https://doi.org/10.1057/s41264-022-00156-x
Malaquias, R. F., & Hwang, Y. (2016). An empirical study on trust in mobile banking: A developing country perspective. *Computers in Human Behavior, 54*, 453–461. https://doi.org/10.1016/j.chb.2015.08.039
Martinez, B. M. B. M., & McAndrews, L. E. L. E. (2022). Do you take..? The effect of mobile payment solutions on use intention: An application of UTAUT2. *Journal of Marketing Analytics*. https://doi.org/10.1057/s41270-022-00175-6
Mehrad, D., & Mohammadi, S. (2017). Word of Mouth impact on the adoption of mobile banking in Iran. *Telematics and Informatics, 34*(7), 1351–1363. https://doi.org/10.1016/j.tele.2016.08.009
Merhi, M., Hone, K., & Tarhini, A. (2019). A cross-cultural study of the intention to use mobile banking between Lebanese and British consumers: Extending UTAUT2 with security, privacy and trust. *Technology in Society, 59*. https://doi.org/10.1016/j.techsoc.2019.101151
Mittal, A., Gupta, M. P., Chaturvedi, M., Chansarkar, S. R., & Gupta, S. (2021). Cybersecurity Enhancement through Blockchain Training (CEBT)–A serious game approach. *International Journal of Information Management Data Insights, 1*(1), 100001. https://doi.org/10.1016/j.jjimei.2020.100001
Mohd Thas Thaker, H., Subramaniam, N. R., Qoyum, A., & Iqbal Hussain, H. (2022). Cashless society, e-wallets and continuous adoption. *International Journal of Finance and Economics*, 3349–3369. https://doi.org/10.1002/ijfe.2596
Mora, C. (2013). Cultures and Organizations: Software of the Mind Intercultural Cooperation and Its Importance for Survival. *Journal of Media Research, 6*(1), 65. https://e-edu.nbu.bg/pluginfile.php/900222/mod_resource/content/1/G.Hofstede_G.J.Hofstede_M.Minkov-CulturesandOrganizations-SoftwareoftheMind3rd_edition2010.pdf.
Mortimer, G., Neale, L., Hasan, S. F. E., & Dunphy, B. (2015). Investigating the factors influencing the adoption of m-banking: A cross cultural study. *International Journal of Bank Marketing, 33*(4), 545–570. https://doi.org/10.1108/IJBM-07-2014-0100
Mzoughi, N., & M'Sallem, W (2013). Predictors of internet banking adoption: Profiling Tunisian postponers, opponents and rejectors. *International Journal of Bank Marketing, 31*(5), 388–408. https://doi.org/10.1108/IJBM-10-2012-0105
Namahoot, K. S., & Laohavichien, T. (2018). Assessing the intentions to use internet banking: The role of perceived risk and trust as mediating factors. *International Journal of Bank Marketing, 36*(2), 256–276. https://doi.org/10.1108/IJBM-11-2016-0159
Naranjo Zolotov, M., Oliveira, T., & Casteleyn, S. (2018). E-participation adoption models research in the last 17 years: A weight and meta-analytical review. *Computers in Human Behavior, 81*, 350–365. https://doi.org/10.1016/j.chb.2017.12.031
Nathan, R. J. R. J., Setiawan, B., & Quynh, M. N. M. N. (2022). Fintech and Financial Health in Vietnam during the COVID-19 Pandemic: In-Depth Descriptive Analysis. *Journal of Risk and Financial Management*, (3), 15. https://doi.org/10.3390/jrfm15030125
Nations, U. (2023). *Human development reports*. https://hdr.undp.org/data-center/human-development-index#/indicies/HDI.
Neves, C., Oliveira, T., & Santini, F. (2022). Sustainable technologies adoption research: A weight and meta-analysis. *Renewable and Sustainable Energy Reviews, 165* (February), Article 112627. https://doi.org/10.1016/j.rser.2022.112627
Nguyen-Viet, B., & Ngoc Huynh, K. (2021). Empirical analysis of internet banking adoption in Vietnam. *Cogent Economics and Finance, 9*(1). https://doi.org/10.1080/23322039.2021.1992876
Ofori, K. S., Boateng, H., Okoe, A. F., & Gvozdanovic, I. (2017). Examining customers' continuance intentions towards internet banking usage. *Marketing Intelligence and Planning, 35*(6), 756–773. https://doi.org/10.1108/MIP-11-2016-0214
Ojo, A. O., Fawehinmi, O., Ojo, O. T., Arasanmi, C., & Tan, C. N. L. (2022). Consumer usage intention of electronic wallets during the COVID-19 pandemic in Malaysia. *Cogent Business & Management, 9*(1), 2056964. https://doi.org/10.1080/23311975.2022.2056964
Oliveira, T., Faria, M., Thomas, M. A., & Popovič, A. (2014). Extending the understanding of mobile banking adoption: When UTAUT meets TTF and ITM. *International Journal of Information Management, 34*(5), 689–703. https://doi.org/10.1016/j.ijinfomgt.2014.06.004
Oliveira, Tiago, Thomas, M., Baptista, G., & Campos, F. (2016). Mobile payment: Understanding the determinants of customer adoption and intention to recommend the technology. *Computers in Human Behavior, 61*(1), 404–414. https://doi.org/10.1016/j.chb.2016.03.030
Organization, W. I. P.. Global innovation index (GII). https://www.wipo.int/global_innovation_index/en/.
Pal, A., Herath, T., De', R., & Rao, H. R (2020). Contextual facilitators and barriers influencing the continued use of mobile payment services in a developing country: Insights from adopters in India. *Information Technology for Development, 26*(2), 394–420. https://doi.org/10.1080/02681102.2019.1701969
Pal, A., Herath, T., De', R., & Raghav Rao, H. (2021). Why do people use mobile payment technologies and why would they continue? An examination and implications from India. *Research Policy, 50*(6). https://doi.org/10.1016/j.respol.2021.104228
Park, J., Ahn, J., Thavisay, T., & Ren, T. (2019). Examining the role of anxiety and social influence in multi-benefits of mobile payment service. *Journal of Retailing and Consumer Services, 47*, 140–149. https://doi.org/10.1016/j.jretconser.2018.11.015
Poromatikul, C., De Maeyer, P., Leelapanyalert, K., & Zaby, S. (2020). Drivers of continuance intention with mobile banking apps. *International Journal of Bank Marketing, 38*(1), 242–262. https://doi.org/10.1108/IJBM-08-2018-0224
Rafdinal, W., & Senalasari, W. (2021). Predicting the adoption of mobile payment applications during the COVID-19 pandemic. *International Journal of Bank Marketing, 39*(6), 984–1002. https://doi.org/10.1108/IJBM-10-2020-0532
Rahadi, R. A. R. A., Nainggolan, Y. A. Y. A., Afgani, K. F. K. F., Yusliza, M. Y. M. Y., Faezah, J. N. J. N., Ramayah, T., Saputra, J., Muhammad, Z., Farooq, K., & Angelina, C. (2022). Towards a cashless society: Use of electronic payment devices among generation z. *International Journal of Data and Network Science, 6*(1), 137–146. https://doi.org/10.5267/J.IJDNS.2021.9.014
Rahi, S., & Abd Ghani, M. (2019). Integration of expectation confirmation theory and self-determination theory in internet banking continuance intention. *Journal of Science and Technology Policy Management, 10*(3), 533–550. https://doi.org/10.1108/JSTPM-06-2018-0057
Rahi, S., Abd Ghani, M., & Hafaz Ngah, A. (2019). Integration of unified theory of acceptance and use of technology in internet banking adoption setting: Evidence from Pakistan. *Technology in Society, 58*, Article 101120. https://doi.org/10.1016/j.techsoc.2019.03.003
Rahman, M., Ismail, I., & Bahri, S. (2020). Analysing consumer adoption of cashless payment in Malaysia. *Digital Business, 1*(1), Article 100004. https://doi.org/10.1016/j.digbus.2021.100004
Rahman, T., Noh, M., Kim, Y. S. Y. S., & Lee, C. K. C. K. (2022). Effect of word of mouth on m-payment service adoption: A developing country case study. *Information Development, 38*(2), 268–285. https://doi.org/10.1177/0266666921999702
Rejman Petrović, D., Nedeljković, I., & Marinković, V (2022). The role of the hedonistic and utilitarian quality dimensions in enhancing user satisfaction in mobile banking. *International Journal of Bank Marketing*. https://doi.org/10.1108/IJBM-03-2022-0112
Riquelme, H. E., & Rios, R. E. (2010). The moderating effect of gender in the adoption of mobile banking. *International Journal of Bank Marketing, 28*(5), 328–341. https://doi.org/10.1108/02652321011064872
Rosario, A. B., Sotgiu, F., De Valck, K., & Bijmolt, T. H. A. (2016). The effect of electronic word of mouth on sales: A meta-analytic review of platform, product, and metric factors. *Journal of Marketing Research, 53*(3), 297–318. https://doi.org/10.1509/jmr.14.0380
Rosenthal, R., & Rubin, D. B. (1991). Further issues in effect size estimation for one-sample multiple-choice-type data. *Psychological Bulletin, 109*(2), 351–352. https://doi.org/10.1037/0033-2909.109.2.351
Roth, P. L., Le, H., Oh, I. S., Van Iddekinge, C. H., & Bobko, P. (2018). Using beta coefficients to impute missing correlations in meta-analysis research: Reasons for caution. *Journal of Applied Psychology, 103*(6), 644–658. https://doi.org/10.1037/apl0000293
Roy, S. K., Balaji, M. S., Kesharwani, A., & Sekhon, H. (2017). Predicting Internet banking adoption in India: A perceived risk perspective. *Journal of Strategic Marketing, 25*(5–6), 418–438. https://doi.org/10.1080/0965254X.2016.1148771
Sampaio, C. H., Ladeira, W. J., & Santini, F. D. O. (2017). Apps for mobile banking and customer satisfaction: A cross-cultural study. *International Journal of Bank Marketing, 35*(7), 1131–1151. https://doi.org/10.1108/IJBM-09-2015-0146
Sarker, S., Chatterjee, S., Xiao, X., & Elbanna, A. (2019). The sociotechnical axis of cohesion for the IS discipline: Its historical legacy and its continued relevance. *MIS Quarterly, 43*(3), 695–719. https://doi.org/10.25300/MISQ/2019/13747
Schmidt, F. L., & Hunter, J. E. (2016). Methods of meta-analysis: Correcting error and bias in research findings. *Methods of meta-analysis: Correcting error and bias in research findings*. https://doi.org/10.4135/9781483398105
Shankar, A., & Jebarajakirthy, C. (2019). The influence of e-banking service quality on customer loyalty: A moderated mediation approach. *International Journal of Bank Marketing, 37*(5), 1119–1142. https://doi.org/10.1108/IJBM-03-2018-0063
Sharma, S., & Khurana, M. K. M. K. (2022). An era of digitalization: Mobile banking adoption in India. *Journal of Science and Technology Policy Management*. https://doi.org/10.1108/jstpm-02-2022-0028
Shaw, N. (2014). The mediating influence of trust in the adoption of the mobile wallet. *Journal of Retailing and Consumer Services, 21*(4), 449–459. https://doi.org/10.1016/j.jretconser.2014.03.008
Singh, N., & Sinha, N. (2020). How perceived trust mediates merchant's intention to use a mobile wallet technology. *Journal of Retailing and Consumer Services, 52*, Article 101894. https://doi.org/10.1016/j.jretconser.2019.101894
Singh, N., Sinha, N., & Liébana-Cabanillas, F. J. (2020). Determining factors in the adoption and recommendation of mobile wallet services in India: Analysis of the effect of innovativeness, stress to use and social influence. *International Journal of Information Management, 50*, 191–205. https://doi.org/10.1016/j.ijinfomgt.2019.05.022
Singh, V., Chen, S., Singhania, M., Nanavati, B., & Gupta, A. (2022). How are reinforcement learning and deep learning algorithms used for big data based decision making in financial industries – A review and research agenda. *International Journal of Information Management Data Insights, 2*(2), Article 100094. https://doi.org/10.1016/j.jjimei.2022.100094
Talwar, S., Dhir, A., Khalil, A., Mohan, G., & Islam, N. (2020). Point of adoption and beyond. Initial trust and mobile-payment continuation intention. *Journal of Retailing and Consumer Services, 55*, Article 102086. https://doi.org/10.1016/j.jretconser.2020.102086
Tam, C., & Oliveira, T. (2016). Understanding the impact of m-banking on individual performance: DeLone & McLean and TTF perspective. *Computers in Human Behavior, 61*, 233–244. https://doi.org/10.1016/j.chb.2016.03.016
Tam, C., & Oliveira, T. (2017). Understanding mobile banking individual performance: The DeLone & McLean model and the moderating effects of individual culture. *Internet Research, 27*(3), 538–562. https://doi.org/10.1108/IntR-05-2016-0117

Tam, C., & Oliveira, T. (2019). Does culture influence m-banking use and individual performance? *Information and Management, 56*(3), 356–363. https://doi.org/10.1016/j.im.2018.07.009

Teoh, W. M.-Y., Chong, S. C., Lin, B., & Chua, J. W. (2013). Factors affecting consumers' perception of electronic payment: An empirical analysis. *Internet Research, 23*(4), 465–485. https://doi.org/10.1108/IntR-09-2012-0199

Tran Le Na, N., & Hien, N. N. (2021). A study of user's m-wallet usage behavior: The role of long-term orientation and perceived value. *Cogent Business and Management, 8*(1). https://doi.org/10.1080/23311975.2021.1899468

Urbach, N., Smolnik, S., & Riempp, G. (2009). The State of Research on Information Systems Success. *Business & Information Systems Engineering, 1*(4), 315–325. https://doi.org/10.1007/s12599-009-0059-y

Venkatesh, V., Thong, J. Y. L., & Xu, X. (2012). Consumer acceptance and use of information technology: Extending the unified theory of acceptance and use of technology. *MIS Quarterly, 36*(1), 157–178.

Verkijika, S. F. (2020). An affective response model for understanding the acceptance of mobile payment systems. *Electronic Commerce Research and Applications, 39*. https://doi.org/10.1016/j.elerap.2019.100905

Webster, J., & Watson, R. T. (2002). Analyzing the past to prepare for the future: Writing a literature review. *MIS Quarterly, 26*(2), xiii–xxiii, 10.1.1.104.6570.

Wood, J. (2008). Methodology for dealing with duplicate study effects in a meta-analysis. *Organizational Research Methods, 11*(1), 79–95. https://doi.org/10.1177/1094428106296638

Yuan, S., Liu, Y., Yao, R., & Liu, J. (2016). An investigation of users' continuance intention towards mobile banking in China. *Information Development, 32*(1), 20–34. https://doi.org/10.1177/0266666914522140

Zhou, Q., Lim, F. J. F. J., Yu, H., Xu, G., Ren, X., Liu, D., Wang, X., Mai, X., & Xu, H. (2021). A study on factors affecting service quality and loyalty intention in mobile banking. *Journal of Retailing and Consumer Services, 60*, Article 102424. https://doi.org/10.1016/j.jretconser.2020.102424

Zhou, T. (2011a). An empirical examination of initial trust in mobile banking. *Internet Research, 21*(5), 527–540. https://doi.org/10.1108/10662241111176353

Zhou, T. (2011b). The effect of initial trust on user adoption of mobile payment. *Information Development, 27*(4), 290–300. https://doi.org/10.1177/0266666911424075

Zhou, T. (2013). An empirical examination of continuance intention of mobile payment services. *Decision Support Systems, 54*(2), 1085–1091. https://doi.org/10.1016/j.dss.2012.10.034

Zhou, Tao, Lu, Y., & Wang, B. (2010). Integrating TTF and UTAUT to explain mobile banking user adoption. *Computers in Human Behavior, 26*(4), 760–767. https://doi.org/10.1016/j.chb.2010.01.013